

Unraveling the Role of Ctla-4 in Intestinal Immune Homeostasis: Insights from a novel Zebrafish Model of Inflammatory Bowel Disease

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This study focuses on the role of a T-cell-specific receptor, *ctla-4*, in a new zebrafish model of IBD-like phenotype. Although implicated in IBD diseases, the function of *ctla-4* has been hard to study in mice as the KO is lethal. *Ctla-4* mutant zebrafish exhibited significant intestinal inflammation and dysbiosis, mirroring the pathology of inflammatory bowel disease (IBD) in mammals, providing a new **valuable** model to the field of IBD research. This is an key study with **convincing** evidence, comprehensive transcriptomic analysis, histological examinations, and functional assays all supporting the findings.

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Abstract

Inflammatory bowel disease (IBD) is a chronic and relapsing immune-mediated disorder characterized by intestinal inflammation and epithelial injury. The underlying causes of IBD are not fully understood, but genetic factors have implicated in genome-wide association studies, including CTLA-4, an essential negative regulator of T cell activation. However, establishing a direct link between CTLA-4 and IBD has been challenging due to the early lethality of CTLA-4 knockout mice. In this study, we identified zebrafish *Ctla-4* homolog and investigated its role in maintaining intestinal immune homeostasis by generating a *Ctla-4*-deficient (*ctla-4*^{-/-}) zebrafish line. These mutant zebrafish exhibited reduced weight, along with impaired epithelial barrier integrity and lymphocytic infiltration in their intestines. Transcriptomics analysis revealed upregulation of inflammation-related genes, disturbing immune system homeostasis. Moreover, single-cell RNA-sequencing analysis indicated increased Th2 cells and interleukin 13 expression, along with decreased innate lymphoid cells and upregulated proinflammatory cytokines. Additionally, *Ctla-4*-deficient zebrafish exhibited reduced diversity and an altered composition of the intestinal microbiota. All these

phenotypes closely resemble those found in mammalian IBD. Lastly, supplementation with Ctl α -4-Ig successfully alleviated intestinal inflammation in these mutants. Altogether, our findings demonstrate the pivotal role of Ctl α -4 in maintaining intestinal homeostasis. Additionally, they offer substantial evidence linking CTLA-4 to IBD and establish a novel zebrafish model for investigating both the pathogenesis and potential treatments.

Introduction

Inflammatory bowel disease (IBD), including Crohn's disease and ulcerative colitis, refers to a group of chronic relapsing inflammation disorders affecting the gastrointestinal tract, that have been increasing in prevalence worldwide [1]. The precise etiology of IBD has yet to be fully elucidated. Conventional epidemiological studies have indicated that IBD tends to run in families and is linked to genetic factors [2, 3]. However, research also suggested that susceptibility gene patterns differ significantly among various geographic populations. Current evidence points towards a complicated interaction involving host genetics, disrupted intestinal microbiota, environmental triggers, and abnormal immune responses [4–6]. Advances in genomic sequencing techniques have allowed for the identification of genetic variants associated with an increased risk of developing IBD. Among these, mutations in immune-related genes have received particular attention. Research on humans with Crohn's disease and mouse models of IBD has shown that genetically susceptible individuals exhibit defects in intracellular pattern-recognition receptors (PRRS), such as toll-like receptors (TLR) and nucleotide-binding oligomerization domain (NOD)-like receptors (NLRs), which are responsible for initiating innate immune responses to eliminate harmful bacteria [7, 8]. Genetic variations in the tumor necrosis factor ligand superfamily member 15 (TNFSF15) and interleukin 23 receptor (IL23R) genes, both involved in suppressing inflammation, have been associated with an increased risk of developing Crohn's disease [9, 10].

Cytotoxic T lymphocyte antigen-4 (CTLA-4), also known as CD152, is one of the most well-established immune checkpoint molecules expressed predominantly on T cells [11–13]. It primarily regulates the early stages of T-cell activation by attenuating downstream signaling of the T cell receptor (TCR) [14–16]. Specifically, CTLA-4 has a much higher affinity for CD80 and CD86 ligands compared to the co-stimulatory receptor CD28 [17, 18]. By outcompeting CD28 for ligand binding, CTLA-4 provides an inhibitory signal that impacts immunological synapse formation and inhibits T-cell proliferation and activation [16, 19]. The immunomodulatory role of CTLA-4 in maintaining immune homeostasis is highlighted through CTLA-4 knockout studies. Germline CTLA-4-deficient is lethal for mice within 3 to 4 weeks due to massive T lymphocyte proliferation and the release of inflammatory cytokines [20–22]. Compared to wild-type T lymphocytes, CTLA-4-deficient T lymphocytes exhibit accelerated development of Th2 cells, leading to significantly enhanced secretion of IL-4 and IL-5 [23, 24]. Additionally, conditional deletion of CTLA-4 in adult mice results in rapid immune activation, multiorgan lymphocyte infiltration, and autoantibody production [25]. Moreover, a selective deficiency of CTLA-4 in Treg cells is sufficient to induce lymphoproliferation and autoimmune diseases in mice [26]. Similarly, associations between polymorphic alleles of CTLA-4 and IBD in humans have been reported in multiple studies [27–29]. Moreover, CTLA-4 is an intriguing target for novel immune checkpoint blockade therapies in cancer treatment, while intestinal inflammation is a common side effect in these clinical trials [30, 31]. Establishing a direct causal relationship between CTLA-4 and IBD has been challenging due to difficulties in finding appropriate models. The early lethality observed in CTLA-4-deficient mice added another layer of complexity to this issue. Zebrafish is a powerful model system for immunological and biomedical research, due to its versatility and high degree of conservation in innate and adaptive immunities [32, 33]. In our current study, we identified the Ctl α -4 homology in zebrafish and successfully developed an adult vertebrate model with homozygous knockout of the *ctla-4* gene for the first time. These *ctla-4*-deficient (*ctla-4*^{-/-}) zebrafish survive but exhibit attenuated growth and weight loss. Notably, *ctla-4*

deficiency leads to an IBD-like phenotype in zebrafish characterized by altered intestinal epithelial cells morphology, abnormal inflammatory response, defects in microbial stratification and composition. Mechanistically, Ctla-4 exerts its inhibitory function by competing with Cd28 for binding to Cd80/86. These findings establish the *ctla-4* knockout zebrafish as an innovative platform to elucidate CTLA-4 immunobiology, model human IBD, and develop novel therapeutic modalities.

Results

Identification of zebrafish Ctla-4

Through a homology search in the NCBI database, we identified the *ctla-4* gene (XM_005167519.4) on zebrafish chromosome 9, which exhibits an exon organization comparable to that of the human CTLA-4 gene (Fig. S1A–C). Zebrafish Ctla-4 is predicted to be a type I transmembrane protein with a molecular weight of approximately 33 kDa, featuring structural characteristics of the immunoglobulin superfamily, which include an N-terminal signal peptide, a single IgV-like extracellular domain, a transmembrane region, and a cytoplasmic tail (Fig. 1A). Multiple amino acid sequence alignments revealed that Ctla-4 contains a ¹¹³LFPPPY¹¹⁸ motif within the ectodomain and a tyrosine-based ²⁰⁶YVKF²⁰⁹ motif in the distal C-terminal region (Fig. 1A). These motifs closely resemble the MYPPPY and YVKM motifs found in mammalian CTLA-4 homologs, which are essential for binding to CD80/CD86 ligands, as well as molecular internalization and signaling inhibition [14, 34, 35]. The IgV-like domain of Ctla-4 was characterized by two-layer β-sandwich and was conserved between zebrafish and humans (Fig. 1B). In contrast, zebrafish Cd28 features a SYPPPF motif in its extracellular region and a FYIQ motif in its intracellular tail, distinguishing it from Ctla-4. Additionally, zebrafish Ctla-4, similar to its counterparts in other species, carries a conserved extracellular ¹²³GNGT¹²⁶ motif, which is absent in zebrafish Cd28 [36]. This structural distinction further differentiates Ctla-4 from Cd28 (Fig. 1A, Fig. S1D). Consistent with this, phylogenetic analysis showed that Ctla-4 clusters with other known CTLA-4 homologs from different species with high bootstrap probability, whereas zebrafish Cd28 groups separately with other CD28s (Fig. S1C). Structurally, Ctla-4 exists as a dimer, and unlike the intracellular localization of CTLA-4 in mammals, Ctla-4 is found on the cell membrane (Fig. 1C and D). By analyzing the splenic scRNA-seq dataset we recently established [37], Ctla-4 was primarily expressed on the T cells, including the Cd4⁺ T and Cd8⁺ T cells (Fig. 1E). This result was verified by immunofluorescence assays on the splenic leukocytes (Fig. 1F).

Ctla-4 deficiency induces inflammatory bowel disease (IBD)-like phenotype

To further investigate the function of Ctla-4, we generated a *ctla-4*^{-/-} zebrafish line with a 14-base deletion in the second exon of the *ctla-4* gene (Fig. 2A–C). The zebrafish appeared grossly normal in appearance; however, the body weight and size were significantly reduced compared with those of wild-type zebrafish (Fig. 2D and E). Anatomically, the *ctla-4*^{-/-} zebrafish were featured by intestine shortening and splenomegaly, suggesting the occurrence of chronic inflammation in the intestines (Fig. 2F and G). For clarification, we first performed histological analysis on the anterior, mid, and posterior intestine segments using H&E staining. Compared to the wild-type zebrafish, the *ctla-4*^{-/-} fish exhibited significant epithelial hyperplasia in the anterior intestine segment, accompanied with a small amount of mucosal inflammatory cell infiltration (Fig. 2H). Moreover, a noteworthy goblet cell loss, reduction of normal mucins and the accumulation of acidic mucins were also observed in *ctla-4*^{-/-} anterior intestine, as detected through Alcian Blue and Periodic Acid-Schiff (AB-PAS) or PAS staining (Fig. 2I and J, Fig. S3A). A small amount of lymphocytic infiltration and mild epithelial damage occurred in the mid

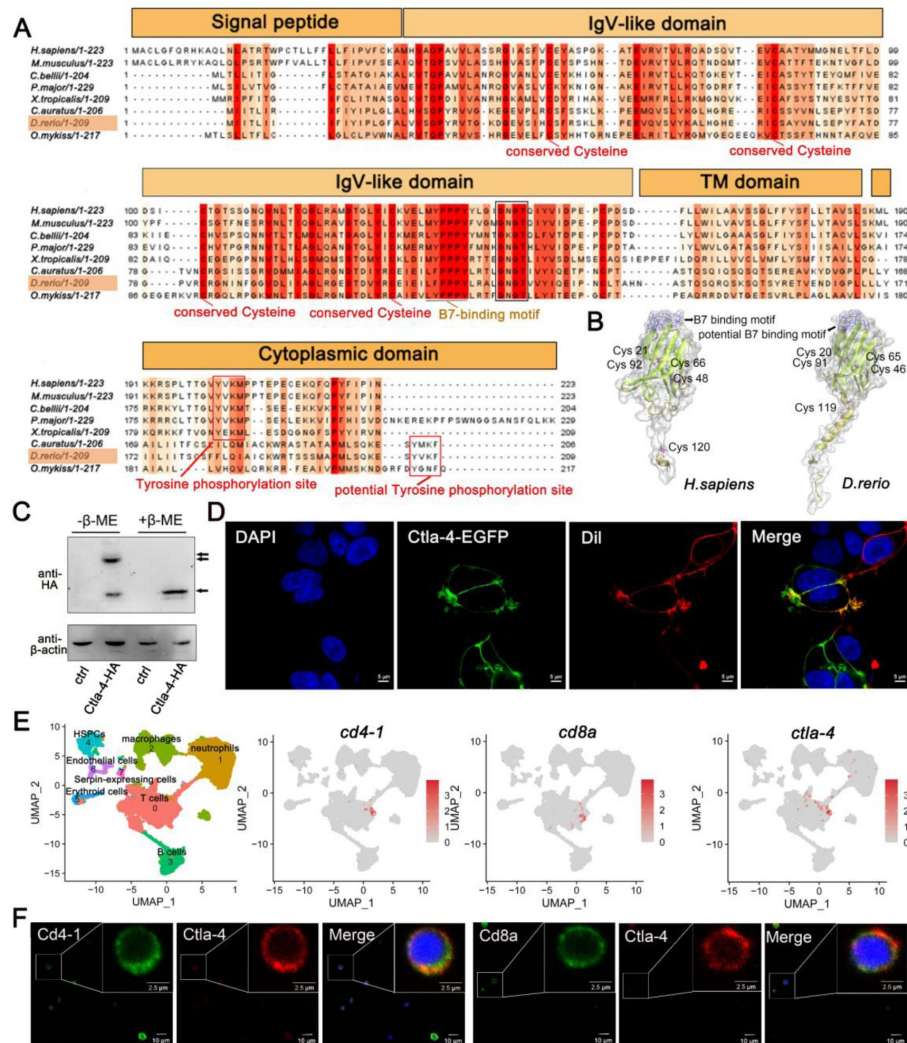


Fig. 1

Characterization of zebrafish Ctla-4.

A Alignment of the Ctla-4 homologs from different species generated with ClustalX and Jalview. The conserved and partially conserved amino acid residues in each species are colored in hues graded from orange to red, respectively. Key features, including conserved cysteine residues, functional motifs, such as B7-binding motif, tyrosine phosphorylation site, and potential tyrosine phosphorylation site, were indicated separately. The signal peptide, IgV-like domain, transmembrane (TM) domain, and cytoplasmic domain were marked above the sequence. **B** The tertiary structure of the zebrafish Ctla-4 ectodomain, as predicted by AlphaFold2, was compared with that of humans. The two pairs of disulfide bonds (Cys²⁰-Cys⁹¹/Cys⁴⁶-Cys⁶⁵ in zebrafish and Cys²¹-Cys⁹²/Cys⁴⁸-Cys⁶⁶ in humans) used to connect the two-layer β -sandwich, and the separate Cys residue (Cys¹¹⁹ in zebrafish and Cys¹²⁰ in humans) involved in the dimerization of the proteins are indicated. Cysteine residues are represented in purple ball-and-stick models, and the identified or potential B7 binding sites are highlighted in blue. **C** Dimer of Ctla-4 was identified by Western blot under reducing (+ β -ME) or non-reducing (- β -ME) conditions. The ctrl represents a control sample derived from cells transfected with an empty plasmid. The monomers and dimers were indicated by single and double arrows, respectively. **D** The subcellular localization of Ctla-4 protein was assessed in HEK293T cells transfected with pEGFPN1-Ctla-4 for 48 hours, imaged using a two-photon laser-scanning microscope (Original magnification, 630 $\times$). Nuclei were stained with DAPI (blue), and cell membranes were stained with DiI (red). **E** UMAP plots showing the relative distribution of common T cell markers (*cd4-1*, *cd8a* and *ctla-4*) based on a splenic single-cell RNA sequencing (scRNA-seq) dataset we recently established [37]. **F** Immunofluorescence staining of lymphocytes isolated from zebrafish blood, spleen, and kidney. Cells were stained with mouse anti-Ctla-4, together with rabbit anti-Cd4-1 or rabbit anti-Cd8a. DAPI stain shows the location of the nuclei. Images were obtained using a two-photon laser-scanning microscope (Original magnification, 630 $\times$).

intestine segment of *ctla-4*^{-/-} zebrafish (Fig. 2H). In posterior intestine of *ctla-4*^{-/-} fish, the intestinal villi were markedly shortened, the epithelial barrier showed severely disrupted, and the intestinal wall became thinner, wherein the mucosal and transmural inflammatory cells were significantly infiltrated (Fig. 2H). Notably, the intestinal lumens in all three intestinal segments were enlarged in the *ctla-4*^{-/-} zebrafish, and the ratio between the length of the intestinal villi and the intestinal ring radius was higher in the *ctla-4*^{-/-} zebrafish intestines compared to those in wild-type zebrafish (Figure S3B). In addition, the ultrastructure analysis revealed that the epithelial cells of posterior intestine in *ctla-4*^{-/-} zebrafish exhibited alteration in tight junction, the loss of adhesion junctions and desmosomes, and disruption of microvilli (Fig. 2K). All these results strongly indicate that Ctla-4 plays a crucial role in preserving intestinal homeostasis in zebrafish. The intestinal phenotype resulting from Ctla-4 deficiency was similar to IBD in mammals.

Molecular mechanisms of Ctla-4 deficiency-induced IBD-like phenotype

To explore the potential molecular mechanisms of Ctla-4 deficiency-induced IBD-like phenotype, we performed transcriptome profiling analysis of intestines from wild-type and *ctla-4*^{-/-} zebrafish. We identified a total of 1,140 differentially expressed genes (DEGs), among which 714 genes were up-regulated, and 426 genes were down-regulated in *ctla-4*^{-/-} zebrafish (Fig. 3A and B). GO enrichment analysis showed that DEGs or up-regulated genes in the top 10 enriched biological processes were associated with the immune response and inflammatory response (Fig. 3C and D). Moreover, the KEGG enrichment analyses indicated that the up-regulated DEGs are primarily involved in the process of cytokine-cytokine receptor interaction and cell adhesion molecules, which are also related to inflammation (Fig. S4A); however, the down-regulated DEGs were significantly enriched in the process of metabolism (Fig. S4B). The intestines of *ctla-4*^{-/-} zebrafish showed significant upregulation of *il1b*, *tnfa*, myeloid-specific peroxidase (*mpx*), matrix metalloproteinase 9 (*mmp9*), chemokine (C-X-C motif) ligand 8a (*cxcl8a*), and *il13*. In contrast, *il10*, a potent anti-inflammatory cytokine, was markedly down-regulated in Ctla-4-deficient intestines (Fig. 3E). The transcriptional change of these genes was confirmed by RT-qPCR (Fig. 3F). By constructing the protein-protein interaction (PPI) network, we found that *il1b* was a major cytokine that played a hub role in promoting the bowel inflammation of *ctla-4*^{-/-} zebrafish (Fig. 3G). Moreover, Gene set enrichment analysis (GSEA) showed that genes involved in the lymphocyte chemotaxis, positive regulation of ERK1 and ERK2 cascade, Calcium and MAPK signaling pathways were positively enriched in *ctla-4*^{-/-} zebrafish intestines, implying a sensitized or activated state of lymphocytes due to the absence of Ctla-4 (Fig. S4C and D). Notably, biological processes related to neutrophil activation and chemotaxis were significantly enriched (Fig. 3C and D). Studies have shown that neutrophils can induce histopathological effects through releasing matrix metalloproteinases (MMPs), neutrophil elastase, and myeloperoxidase (MPO) [38]. To confirm the association between neutrophils and Ctla-4-deficient intestinal inflammation, the MPO level was examined. As a support, MPO activity was markedly increased in the intestines and peripheral blood of *ctla-4*^{-/-} zebrafish (Fig. 3H). Besides, a number of biological markers or susceptibility genes of IBD observed in mammals, including c-reactive protein 6 (*crp6*), *crp7*, MMPs, haptoglobin, *il23r*, insulin-like growth factor binding protein 1 a (*igfbp1a*), cAMP responsive element modulator b (*cremb*) and lymphocyte specific protein 1 b (*lsp1b*), were highly expressed in the *ctla-4*^{-/-} zebrafish (Fig. 3I and J) [9, 39, 40], suggesting the presence of a conserved molecular network underlying IBD pathogenesis between *ctla-4*^{-/-} zebrafish and mammalian models.

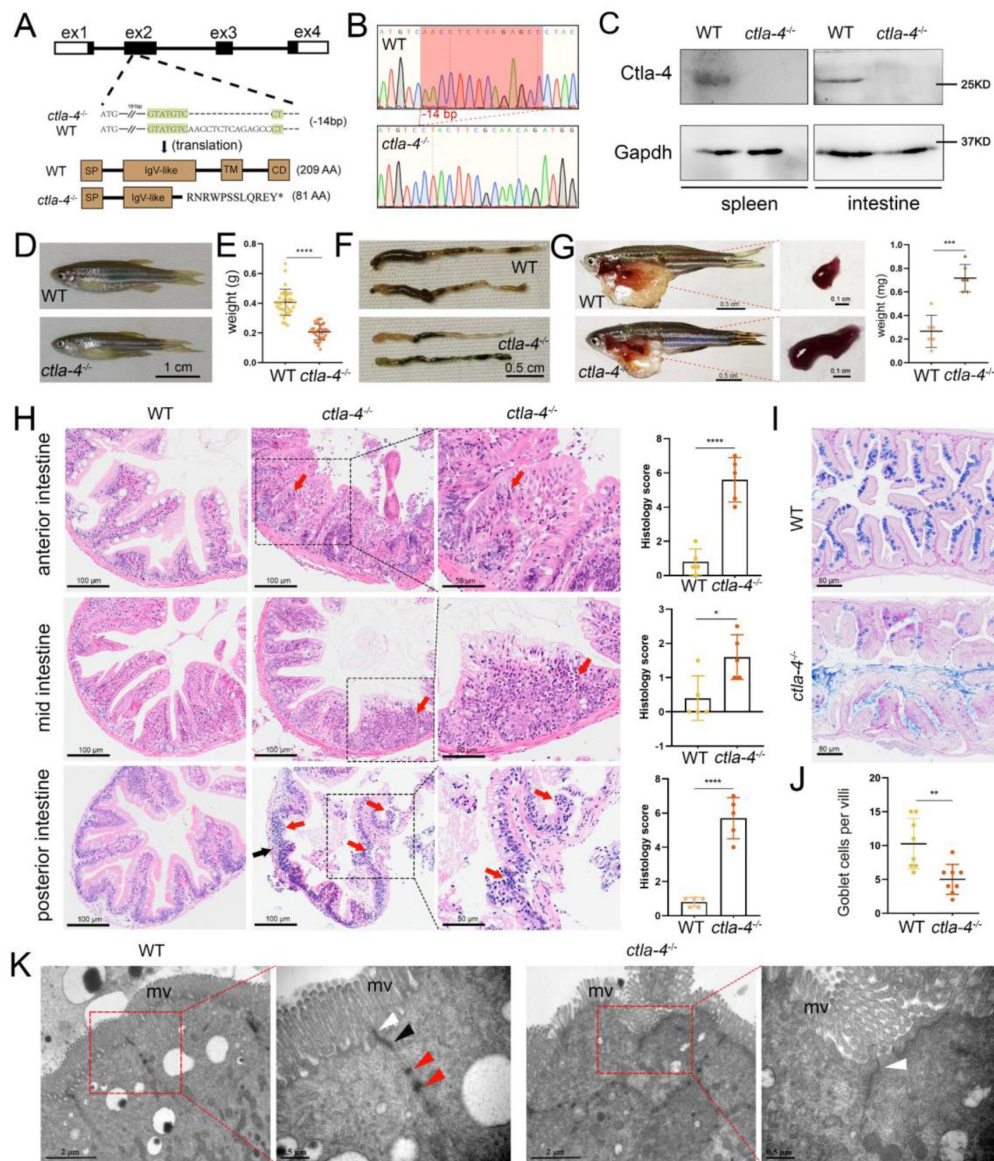


Fig. 2

Examination on the IBD-like phenotype in *ctla-4*^{-/-} zebrafish.

A Generation of a homozygous *ctla-4*-deficient (*ctla-4*^{-/-}) zebrafish line through CRISPR/Cas9-based knockout of *ctla-4* gene on chromosome 9. A 14-bp deletion mutation in exon 2 results in a premature stop at codon 82, which is predicted to produce a defective Ctla-4 protein containing 81 amino acids. **B** Genotyping of the deficiency of *ctla-4* gene by Sanger sequencing. **C** Knockout efficiency of Ctla-4 selectively examined in spleen and gut tissues of *ctla-4*^{-/-} zebrafish by Western blot analysis. Gapdh serves as a loading control. **D** Normal gross appearance of adult wild-type (WT) and *ctla-4*^{-/-} zebrafish. **E** Body weight statistics of WT and *ctla-4*^{-/-} zebrafish (n = 30). **F** The change of intestine length in WT and *ctla-4*^{-/-} zebrafish. **G** The change of splenic size in WT and *ctla-4*^{-/-} zebrafish. **H** Representative H&E staining analysis of histopathological changes and quantitation of histology scores in the anterior, mid and posterior intestines from WT and *ctla-4*^{-/-} zebrafish. Red arrows denote mucosal inflammatory cell infiltration, and black arrow indicates transmural inflammatory cell infiltration. **I** AB-PAS staining was used to analyze the mucin components and the number of goblet cells in anterior intestine from WT and *ctla-4*^{-/-} zebrafish (n = 5). **J** Quantitation analysis of goblet cells of each villus in the foregut of WT and *ctla-4*^{-/-} zebrafish (n = 8). **K** Observation of cell junctions between intestinal epithelial cells in posterior intestines from WT and *ctla-4*^{-/-} zebrafish under TEM (Hitachi Model H-7650). White triangles indicate tight junctions, black triangle indicates adhesion junctions, and red triangles indicate desmosomes. Data are presented as mean ± standard deviation (SD). Statistical significance was assessed through an unpaired Student's t test (*p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001).

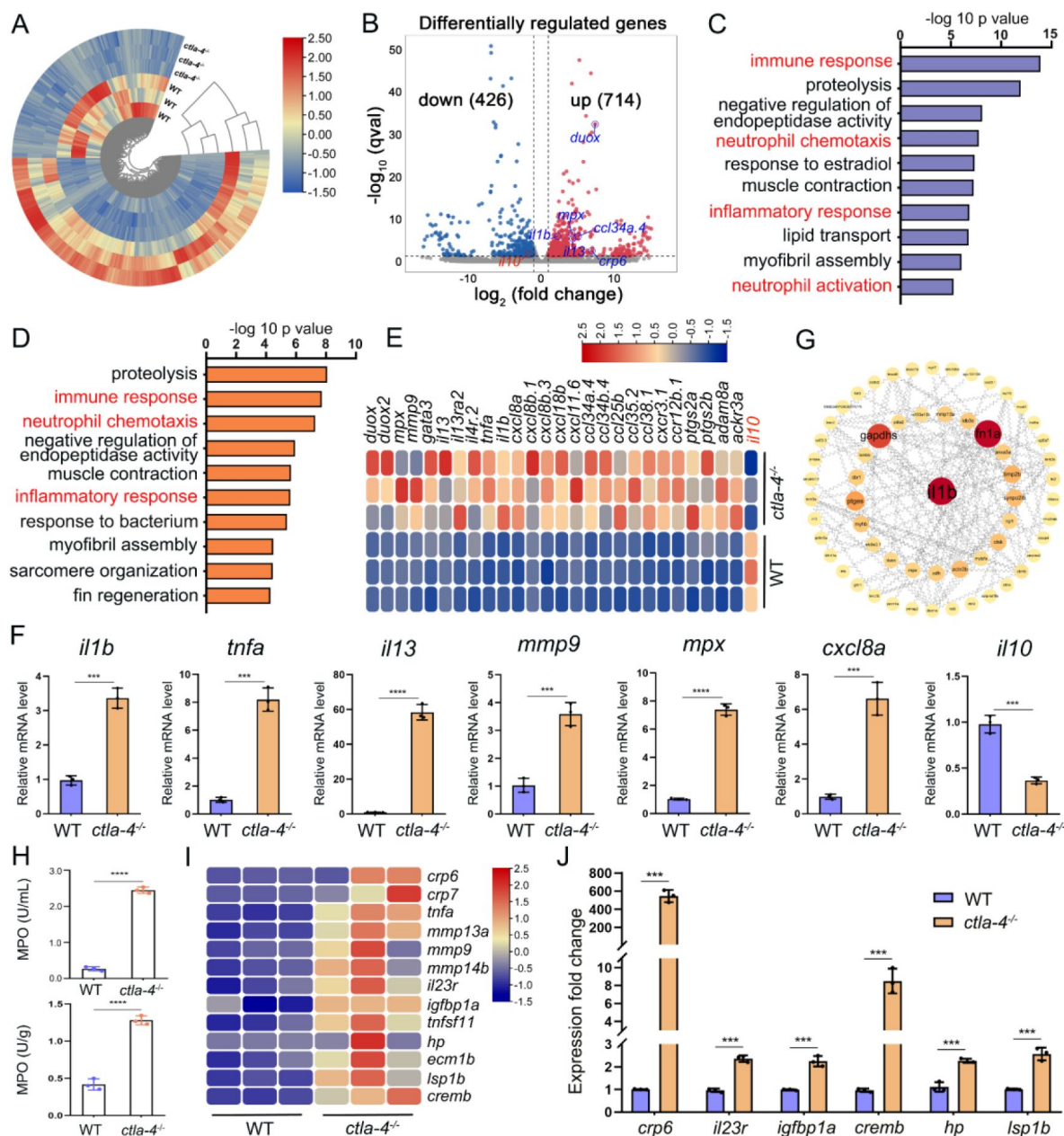


Fig. 3

RNA-sequencing analysis of the molecular implications associated with the IBD-like phenotype in *ctla-4*^{-/-} zebrafish.

A Heatmap of different expressed genes between the intestines from wild-type (WT) and *ctla-4*^{-/-} zebrafish. **B** Volcano plot showing the up-/down-regulated genes in the intestines of *ctla-4*^{-/-} zebrafish compared with those of WT zebrafish. Red represents up-regulated genes, while blue denotes down-regulated genes. **C** GO analysis showing top 10 terms in biological processes of DEGs. **D** GO analysis showing top 10 terms in biological processes of all up-regulated genes. **E** Heatmap showing row-scaled expression of the representative differently expressed inflammation and chemotaxis-related genes. **F** The mRNA expression levels of important genes associated with inflammation and chemokines confirmed by real-time qPCR. **G** Protein-protein interaction network was constructed using the DEGs. The nodes represent the proteins (genes); the edges represent the interaction of proteins (genes). **H** The MPO activity in the intestines (up) and peripheral blood (down). **I** Heatmap showing row-scaled expression of the IBD biomarker genes and IBD-related genes. **J** The mRNA expression levels of representative IBD biomarker genes and IBD-related genes were analyzed by real-time qPCR. Data are presented as mean $\pm$ standard deviation (SD). Statistical significance was assessed through an unpaired Student's t test (** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$).

Cellular mechanisms of *Ctla-4* deficiency-induced IBD-like phenotype

To investigate the cellular mechanisms underlying the IBD-like phenotype induced by *Ctla-4* deficiency, we performed scRNA-seq analysis on intestines from wild-type and *ctla-4*^{-/-} zebrafish using the 10× Genomics Chromium platform. We obtained nine discrete clusters from 7,539 cells of wild-type and *ctla-4*^{-/-} zebrafish (Fig. 4A). These clusters of cells were classified as enterocytes, enteroendocrine cells, smooth muscle cells, neutrophils, macrophages, B cells, and a group of T/NK/ILC-like cells based on their co-expression of lineage marker genes (Fig. 4B and C, Fig. S6A and B). Due to severe epithelial disruption and inflammatory cell infiltration in *ctla-4*^{-/-} zebrafish intestines, we focused on the pathological process and immune reactions in enterocytes and immune cell populations. KEGG analysis showed that apoptotic pathway was highly enriched in enterocytes of *ctla-4*^{-/-} zebrafish, suggesting that aberrant apoptosis contributes to the epithelial injury (Fig. S6C). Subsequently, we conducted a TUNEL assay to detect apoptosis in the posterior intestines from both wild-type and *ctla-4*^{-/-} zebrafish. The results showed a higher number of apoptotic cells in the intestines of *ctla-4*^{-/-} zebrafish (Fig. S6D). Additionally, genes functionally involved in the formation of tight and adhesion junctions, such as *oclna*, *cdh1*, *pcdh1b* and *cldn15a*, were significantly down-regulated in enterocytes of *ctla-4*^{-/-} zebrafish (Fig. 4D), consistent with the pathological observation under electron microscope. Furthermore, inflammation-related genes and pathways were significantly up-regulated and enriched in neutrophils, B cells, and macrophages of *ctla-4*^{-/-} zebrafish, suggesting active inflammatory responses (Fig. 4E-G, Fig. S6E). By sub-clustering analysis, six subpopulations were classified from T/NK/ILC-like cell groups based on their expression with a set of marker genes. These subpopulations include Cd8⁺ T cells, ILC3-like cells, maturing Ccr7^{high} T cells, NKT-like cells, and two groups of Th2 cells (Fig. 5A-C, Fig. S7A). The abundances of NKT-like and two subsets of Th2 cells were significantly increased in the intestines of *ctla-4*^{-/-} zebrafish (Fig. 5D-F). These findings were further validated by RT-qPCR detection of their corresponding marker genes (Fig. S7B and C). These cells exhibited high expression levels of *il13* (Fig. 5G and H). Specifically, the second subset of Th2 cells was seldom observed in the intestine of wild-type zebrafish, indicating their unique role in the pathogenesis of IBD-like phenotype in *ctla-4*^{-/-} zebrafish (Fig. 5D-F). KEGG analysis of up-regulated genes from *ctla-4*^{-/-} NKT-like and Th2 cells indicated that *Ctla-4* deficiency is positively associated with the inflammatory cytokine-cytokine receptor interaction, PPAR, calcium and MAPK signaling pathways, cellular adhesion and mucosal immune responses (Fig. 5I and J, Fig. S7D). Although the abundance of Cd8⁺ T cells was not significantly changed in *Ctla-4*-deficient intestines, the inflammatory genes and pathways were up-regulated and enriched in the subset of T cells (Fig. S7E and F). Notably, the proportion of ILC3-like cells was decreased, and they highly expressed *il17a/f1* and *il17a/f3* in the *Ctla-4*-deficient intestines (Fig. 5D-F, and K). Investigations have consistently reported a substantial decline in the population of ILC3s within the inflamed intestines and IL-17A-secreting ILC3s play a significant role in the development of intestinal inflammation [41–46]. Thus, the reduced ILC3-like cells and increased expression of *il17a/f1* and *il17a/f3* may be responsible for intestinal inflammation induced by *Ctla-4* deficiency.

Decreased microbiota diversity in *ctla-4*^{-/-} zebrafish intestines

The intestinal microbiota plays a crucial role in host functions such as nutrient acquisition, metabolism, epithelial cell development and immunity. Notably, lower microbiota diversity has consistently been observed in patients with IBD phenotype [47, 48], making it a valuable indicator of host health. Therefore, we further analyzed whether microbes are involved in the *Ctla-4*-deficiency induced intestinal inflammation. The results revealed a significantly higher number of amplicon sequence variants (ASVs) in wild-type zebrafish intestines, with 730 ASVs unique to the wild-type group and 276 ASVs exclusively found in *ctla-4*^{-/-} group (Fig. 6A). Furthermore, the Shannon index and the Simpson index indicated a decreased microbial diversity

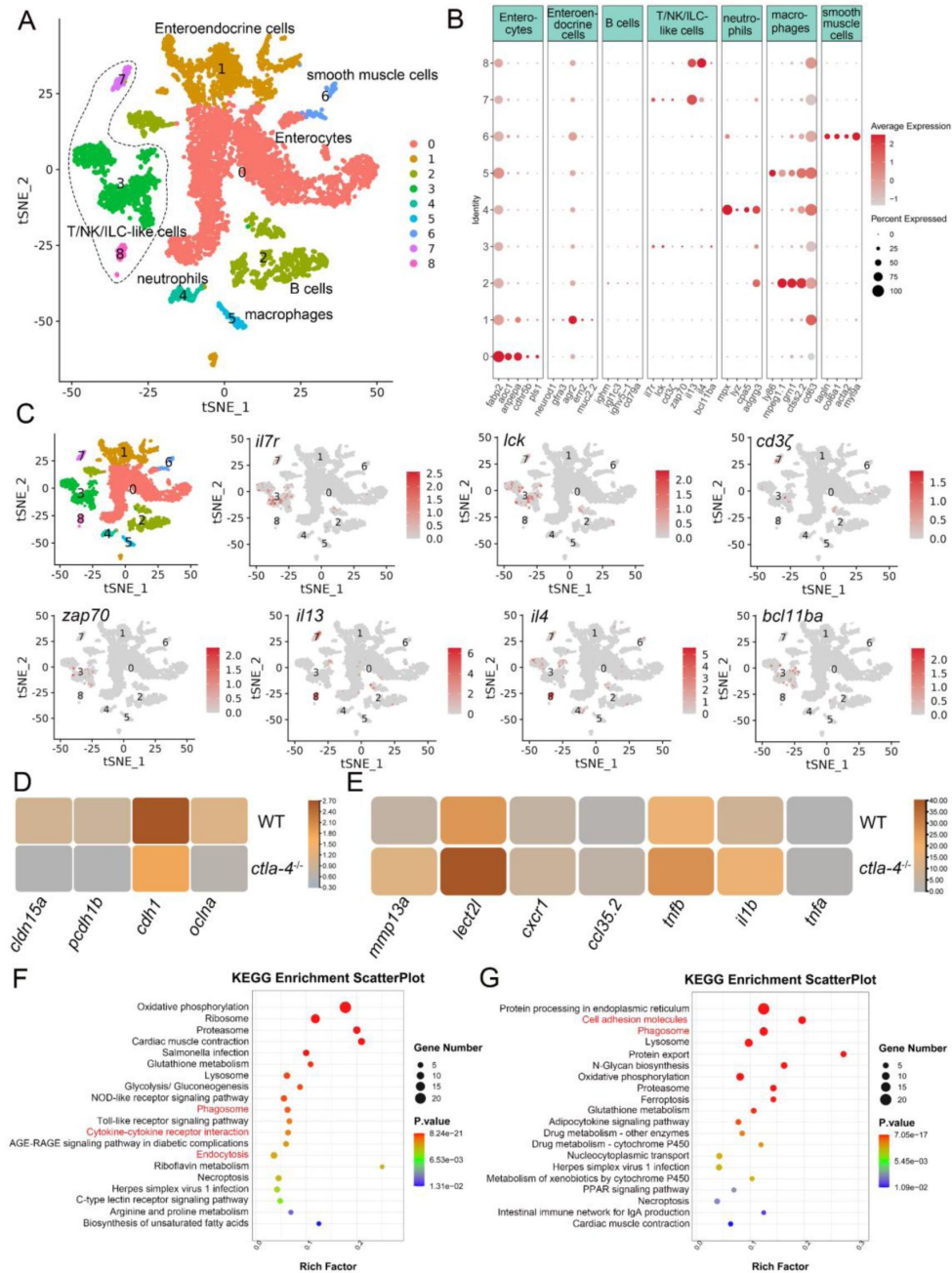


Fig. 4

Single-cell RNA sequencing analysis of the major cell types associated with the IBD-like phenotype in *ctla-4*^{-/-} zebrafish.

A Classification of cell types from zebrafish intestines by tSNE embedding. **B** Dot plot showing the expression levels of lineage marker genes and percentage of cells per cluster that express the gene of interest. **C** Expression maps of T cell associated markers within the cell populations of the zebrafish intestines. **D** Heatmap showing the mean expression levels of genes associated with tight and adhesion junctions in enterocytes across samples from wild-type (WT) and *ctla-4*^{-/-} zebrafish. **E** Heatmap showing the mean expression levels of inflammation-related genes involved in cytokine-cytokine receptor interactions in neutrophils from WT and *ctla-4*^{-/-} zebrafish samples. **F** KEGG enrichment analysis showing the top 15 terms of up-regulated genes in neutrophils from the *ctla-4*^{-/-} sample versus the WT sample. **G** KEGG enrichment analysis showing the top 15 terms of up-regulated genes in macrophages from the *ctla-4*^{-/-} sample versus the WT sample.

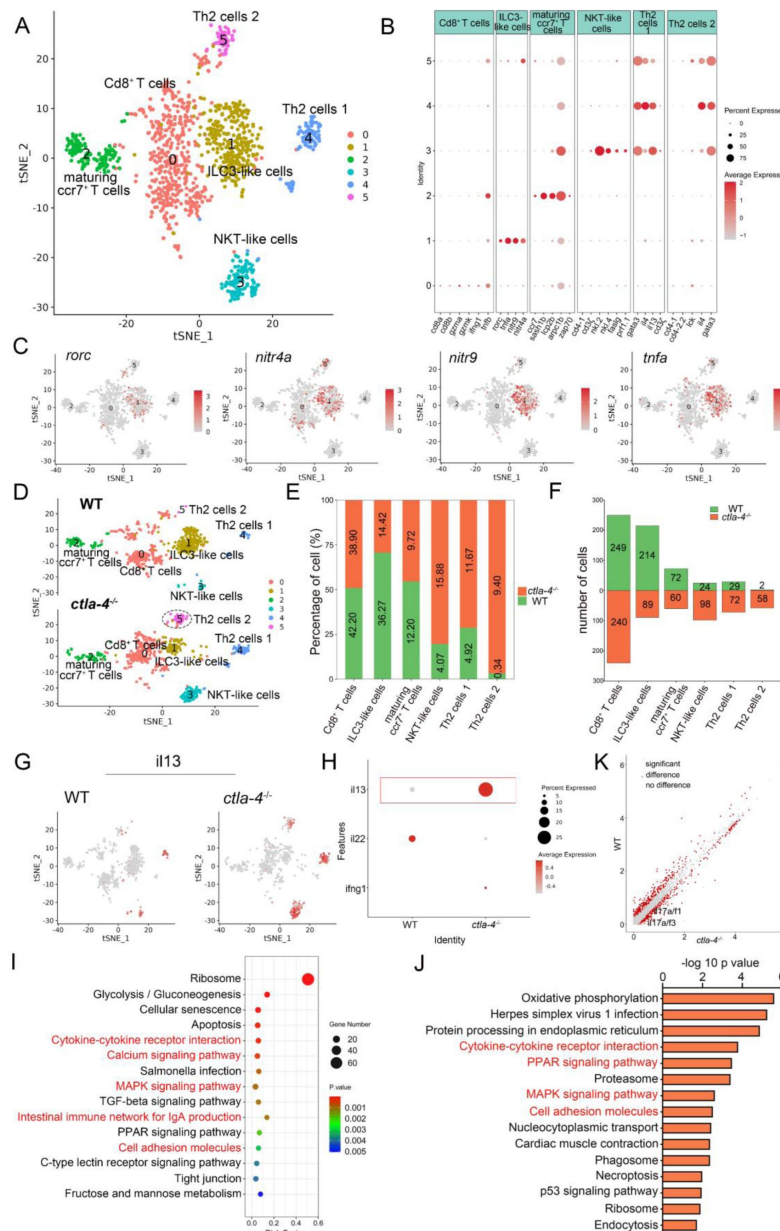


Fig. 5

Single-cell RNA sequencing analysis of the subset immune-cells associated with the IBD-like phenotype in *ctla-4*^{-/-} zebrafish.

A Classification of subset cells from the T/NK/ILC-like group by tSNE embedding. **B** Dot plot showing the mean expression levels of subset marker genes and percentage of cells per cluster that express the gene of interest. **C** Marker gene expression in individual cluster identifying this cluster as ILC3-like cells. **D** Changes in the composition of subset cells between samples from wild-type (WT) and *ctla-4*^{-/-} zebrafish. A significantly increased Th2 subset (referred to as Th2 cells 2) in the *ctla-4*^{-/-} sample was highlighted with a black dashed circle. **E** Histogram showing the different ratios of subset cells between the WT and *ctla-4*^{-/-} samples. **F** Histogram presenting the different numbers of subset cells between the WT and *ctla-4*^{-/-} samples. **G** Mean expression levels of the cytokine *il13* within different subset cells between the WT and *ctla-4*^{-/-} samples. **H** Dot plot illustrating the mean expression of *il13* in T/NK/ILC-like cells from WT and *ctla-4*^{-/-} zebrafish. **I** KEGG enrichment analysis showing the top 15 terms of the Th2 cells 2 genes from *ctla-4*^{-/-} zebrafish. **J** KEGG enrichment analysis showing the top 15 terms of up-regulated genes in NKT-like cells. **K** Scatter plot showing the DEGs of ILC3-like cells in WT and *ctla-4*^{-/-} zebrafish. The *il17a/f1* and *il17a/f3* was shown in the scatter plot.

in *ctla-4*^{-/-} zebrafish intestines (**Fig. 6B**). and the Principal Coordinate Analysis (PCoA) using Bray Curtis distance revealed a significant separation in the microbial composition between *ctla-4*^{-/-} group and the wild-type group (**Fig. 6C**). To gain insights into the microbial community composition, we analyzed the identified microbial populations at the class and family level. Alphaproteobacteria and Gammaproteobacteria were found to be the most prevalent classes. Relative to wild-type group, Ctla-4 deficiency resulted in a significant reduction in Alphaproteobacteria abundance. However, the Gammaproteobacteria, one of the main classes of Gamma-negative pathogenic bacteria expanded under inflammation conditions, was increased, although the change did not reach statistical significance (**Fig. 6D and E**) [49]. In addition, we observed a decreased relative abundance of short-chain fatty acids (SCFAs)-producing Bacilli and Verrucomicrobiae, the latter of which contributes to glucose homeostasis and intestinal health (**Fig. 6F and G**) [50, 51]. Notably, the family-level analysis revealed a notable enrichment of Enterobacteriaceae, overgrowing under host inflammatory conditions, and the Shewanellaceae, serving as the most important secondary or opportunistic pathogens, in *ctla-4*^{-/-} zebrafish (**Fig. 6H and I**). To identify differentially abundant bacterial taxa between the wild-type and *ctla-4*^{-/-} zebrafish, we conducted Linear discriminant analysis (LDA) effect size (LEfSe). The results showed that several potentially opportunistic pathogens, including Gammaproteobacteria, Enterobacterales, and Aeromonadales were found to be overrepresented in *ctla-4*^{-/-} zebrafish (**Fig. 6J**). In contrast, Actinobacteriota, Cetobacterium, and Planctomycetota (Planctomycetes) were more abundant in the wild-type zebrafish. These findings strongly indicated an association between Ctla-4 deficiency-induced gut inflammation and dysbiosis, as characterized by decreased microbial diversity, loss of potentially beneficial bacteria, and expansion of pathobionts.

Inhibitory role of Ctla-4 in T cell activation

Given that Ctla-4 is primarily expressed on T cells (**Fig. 1E-F**), its absence has been shown to induce intestinal immune dysregulation, indicating a crucial role of this molecule as a conserved immune checkpoint in T cell inhibition. Mechanistically, Ctla-4 may inhibit T cell activation by obstructing the Cd80/86-Cd28 costimulatory pathway, a mechanism conserved in mammalian species. To elucidate the regulatory role of Ctla-4 in costimulatory signal-dependent T cell activation, we conducted a series of blockade and activation assays using anti-Ctla-4 antibody, recombinant soluble Ctla-4-Ig (sCtla-4-Ig), sCd28-Ig, and sCd80/86 proteins in a PHA-stimulating and mixed lymphocyte reaction (MLR) model. In this system, sCtla-4-Ig and sCd28-Ig served as antagonists to block membrane-bound Cd80/86, while sCd80/86 acted as an agonist for membrane-bound Cd28 (**Fig. S8A-C**). As expected, the proliferation of lymphocytes from *ctla-4*^{-/-} zebrafish was more pronounced compared to wild-type controls, and the addition of sCtla-4-Ig effectively suppressed this proliferation (**Fig. 7A and B**). These findings indicate that the absence of Ctla-4 leads to enhanced lymphocyte activation, which can be counteracted by sCtla-4 administration, underscoring the inhibitory function of Ctla-4 in T cell regulation. Consistent with these results, blockade of Ctla-4 using anti-Ctla-4 Ab significantly promoted the proliferation of lymphocytes from wild-type zebrafish (**Fig. 7C**). Furthermore, sCd28-Ig administration inhibited the proliferation of lymphocytes from *ctla-4*^{-/-} zebrafish (**Fig. 7D**), whereas sCd80/86 promoted the expansion of Ctla-4-deficient lymphocytes (**Fig. 7E**). Based on these results, we concluded that the presence of Ctla-4 obstructs the Cd80/86-Cd28 mediated costimulatory signaling, consequently impeding cell proliferation. To further investigate the molecular interactions between Cd28, Ctla-4, and Cd80/86, we employed AlphaFold2 to predict the structures of Cd80/86-Cd28 and Cd80/86-Ctla-4 complexes. A total of 25 models were generated for each complex and subsequently aligned with Cd80/86. The predictions indicated that both Cd28 and Ctla-4 form a well-defined interface with Cd80/86, utilizing the same binding site (**Fig. S8D** and **E**). This well-defined interface was corroborated by lower predicted aligned error (PAE) scores for each model, as marked by the red dashed square (**Fig. 7F and G**). Subsequently, co-immunoprecipitation (Co-IP) assays were conducted to provide compelling evidence for the molecular interactions between Cd80/86 and Cd28 or Ctla-4. Flow cytometry analysis further revealed dose-dependent associations between Cd80/86 and Cd28 or Ctla-4 in HEK293T cells (**Fig. 7H**, S8F). Additionally, microscale

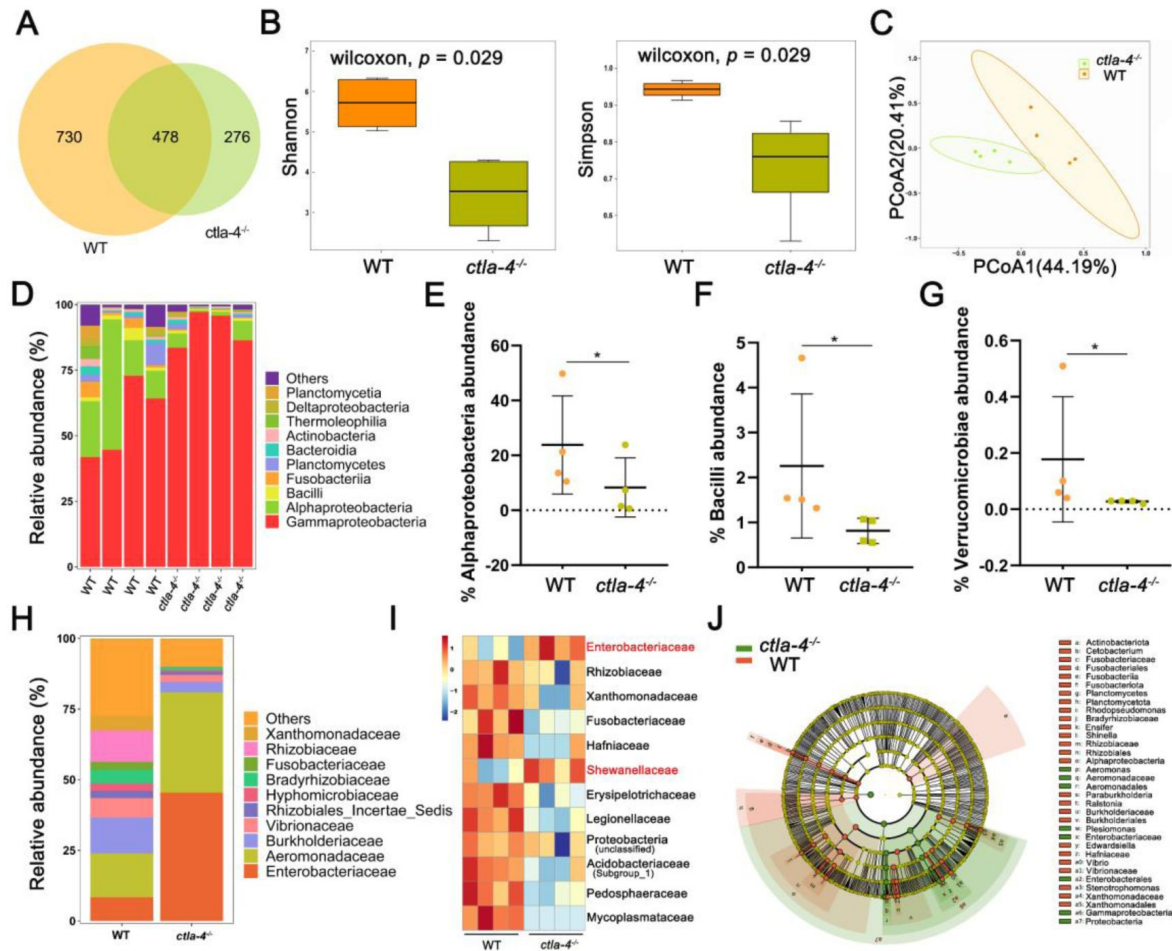


Fig. 6

Alteration in microbial composition in the intestines of *ctla-4^{-/-}* zebrafish.

A Venn diagram showing the number of ASVs in zebrafish intestinal microbiota. **B** Alpha-diversity of microbes was calculated through Shannon index and Simpson index. **C** Beta-diversity analyzed based on PCoA was shown by using Bray Curtis distance. **D** The relative abundance of intestinal microbiota at the class level. **E-G** The relative abundance of Alphaproteobacteria (**E**), Bacilli (**F**) and verrucomicrobiae (**G**) in the intestines from the wild-type (WT) and *ctla-4^{-/-}* zebrafish. * $p < 0.05$. **H** The relative abundance of intestinal microbiota at the family level. **I** Heatmap showing row-scaled expression of the differential abundances of bacterial communities at family level in the WT and *ctla-4^{-/-}* zebrafish ($p < 0.05$). **J** Cladogram representation of LEfSe analysis showing the differentially abundant bacterial taxa between the intestines from WT (red) and *ctla-4^{-/-}* (green) zebrafish ($p < 0.05$).

thermophoresis assays demonstrated that Ctla-4 exhibits a higher binding affinity for Cd80/86 than Cd28, as evidenced by a lower equilibrium association constant value ($K_D = 0.50 \pm 0.25 \mu\text{M}$ vs. $K_D = 2.64 \pm 0.45 \mu\text{M}$) (Fig. 7I). These findings suggest that Ctla-4 exerts its inhibitory function by competing with Cd28 for binding Cd80/86.

sCtla-4-Ig alleviates IBD-like phenotype

As described above, engagement of Cd80/86 by sCtla-4-Ig effectively suppressed T cell activation *in vitro* (Fig. 7B), indicating that sCtla-4-Ig holds promise as a potential intervention for IBD-like phenotype. This is supported by the observation that Ctla-4-deficient zebrafish treated with sCtla-4-Ig exhibited obvious body weight restoration compared to those treated with the IgG isotype control (Fig. 8A). To provide further evidence, histological analysis was performed on the posterior intestine, which is known to experience severe tissue destruction under Ctla-4 deficient conditions. As expected, Ctla-4-Ig treatment resulted in a significant decrease in lymphocyte infiltration and an improvement in the epithelial barrier (Fig. 8B). Moreover, Ctla-4-Ig treatment significantly reduced the expression of pro-inflammatory genes, including *il13*, *tnfa*, *mpx*, *mmp9*, and *cxcl8a*, as well as *igfbp1a*, *cremb*, and *lsp1a*, which are susceptibility genes for IBD observed in mammals (Fig. 8C and D). These findings demonstrate that the supplementation of Ctla-4-Ig alleviates intestinal inflammation in Ctla-4-deficient zebrafish, highlighting its potential as a therapeutic intervention for CTLA-4 deficiency-induced IBD in mammals.

Discussion

As an essential negative regulator of T cell activation, dysfunction of CTLA-4 was implicated in various diseases in both humans and murine models [20, 21, 52]. Numerous previous studies have established the connection between CTLA-4 and autoimmune thyroiditis, Graves' disease, myocarditis, pancreatitis, multiple sclerosis, rheumatoid arthritis, and type I diabetes [53–59]. However, the involvement of CTLA-4 in IBD has been understudied. Several investigations have reported that haploinsufficiency resulting from mutations in CTLA-4 in humans is associated with IBD, and genome-wide association studies (GWAS) have implicated CTLA-4 as a susceptibility gene for IBD [60–62]. Nevertheless, the exact contributions and mechanisms of CTLA-4 deficiency in the occurrence and pathology of IBD remain incompletely understood, primarily due to the lack of animal models attributable to the lethality of CTLA-4 knockout in mice. In this study, we identified the Ctla-4 homolog in zebrafish, and discovered that defect in Ctla-4 did not have a severe lethal effect, but did show a clear IBD-like phenotype. This makes zebrafish an attractive animal model for investigating the molecular and cellular mechanisms underlying Ctla-4 mediated IBD.

Multiple lines of histopathological evidence demonstrated the IBD-like phenotype in Ctla-4-deficient zebrafish. Key features include epithelial hyperplasia, disruption of epithelial integrity, loss of goblet cells, increased acidic mucus production, intestinal lumen enlargement, inflammatory cell infiltration, and elevated expression of pro-inflammatory cytokines in the inflamed intestines. These characteristics, such as epithelial hyperplasia, goblet cell depletion, inflammatory cell infiltration, and upregulated pro-inflammatory cytokine expression, closely resemble those observed in the dextran sodium sulfate (DSS)-induced IBD model in mice [63]. Similarly, mononuclear cell infiltration and significant upregulation of the $\text{il1}\beta$ cytokine have been reported in the trinitrobenzenesulfonic acid (TNBS)-induced IBD model in adult zebrafish [64]. In zebrafish larvae, the TNBS-induced IBD-like phenotype also exhibits an enlarged intestinal lumen, although goblet cell numbers were increased [65]. Additionally, neutrophilic inflammation and acidic mucin accumulation have been observed in the DSS-induced enterocolitis model in zebrafish larvae [66]. In contrast, the soybean meal-induced enteritis (SBMIE) phenotype in zebrafish larvae shows no significant structural changes in intestinal architecture,

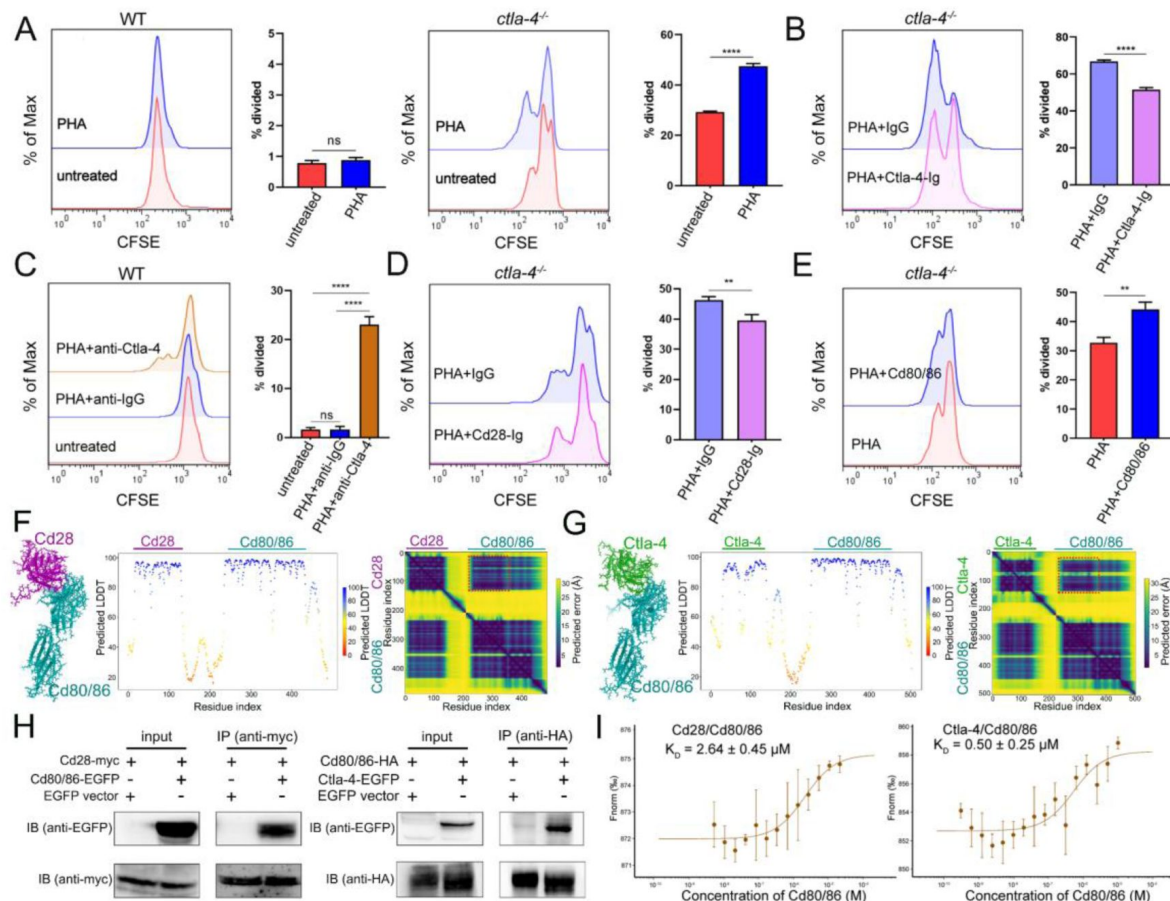


Fig. 7

Examination on the inhibitory function of Ctla-4 in T cell activation.

A Assessment of the proliferative activity of T cells from wild-type (WT) and *ctla-4*^{-/-} zebrafish by a mixed lymphocyte reaction combined with PHA-stimulation. The CFSE dilution, which served as an indicator of lymphocyte proliferation, was measured through flow cytometry. **B** Assessment of the proliferative activity of lymphocytes from *ctla-4*^{-/-} zebrafish by the administration of sCtla-4-Ig. **C** Assessment of the proliferative activity of lymphocytes from WT zebrafish by supplementing anti-Ctla-4 antibody. **D** Assessment of the proliferative activity of lymphocytes from *ctla-4*^{-/-} zebrafish by the administration of sCd28-4-Ig. **E** Assessment of the proliferative activity of lymphocytes from *ctla-4*^{-/-} zebrafish by the administration of recombinant sCd80/86 protein. **F, G** Interactions between Cd80/86 and Cd28 (**F**), and Cd80/86 and Ctla-4 (**G**) as predicted by AlphaFold2. On the left are structural models depicting Cd80/86 in complex with Cd28 or Ctla-4. The center panels display per-residue model confidence scores (pLDDT) for each structure, using a color gradient from 0 to 100, where higher scores indicate increased confidence. The right panels show the predicted aligned error (PAE) scores for each model. The well-defined interfaces between Cd28 or Ctla-4 and Cd80/86 are highlighted with red dashed squares. **H** The interaction between Cd80/86 and Cd28 (left), and Cd80/86 and Ctla-4 (right) were verified by Co-IP. **I** Binding affinities of the Cd80/86 protein for the Cd28 and Ctla-4 proteins were measured by the microscale thermophoresis (MST) assay. The K_D values are provided. Data are presented as mean $\pm$ standard deviation (SD), which were calculated from three independent experiments. Statistical significance was assessed through an unpaired Student's *t* test (***p* < 0.01; ****p* < 0.001; ns denotes no statistical significance).

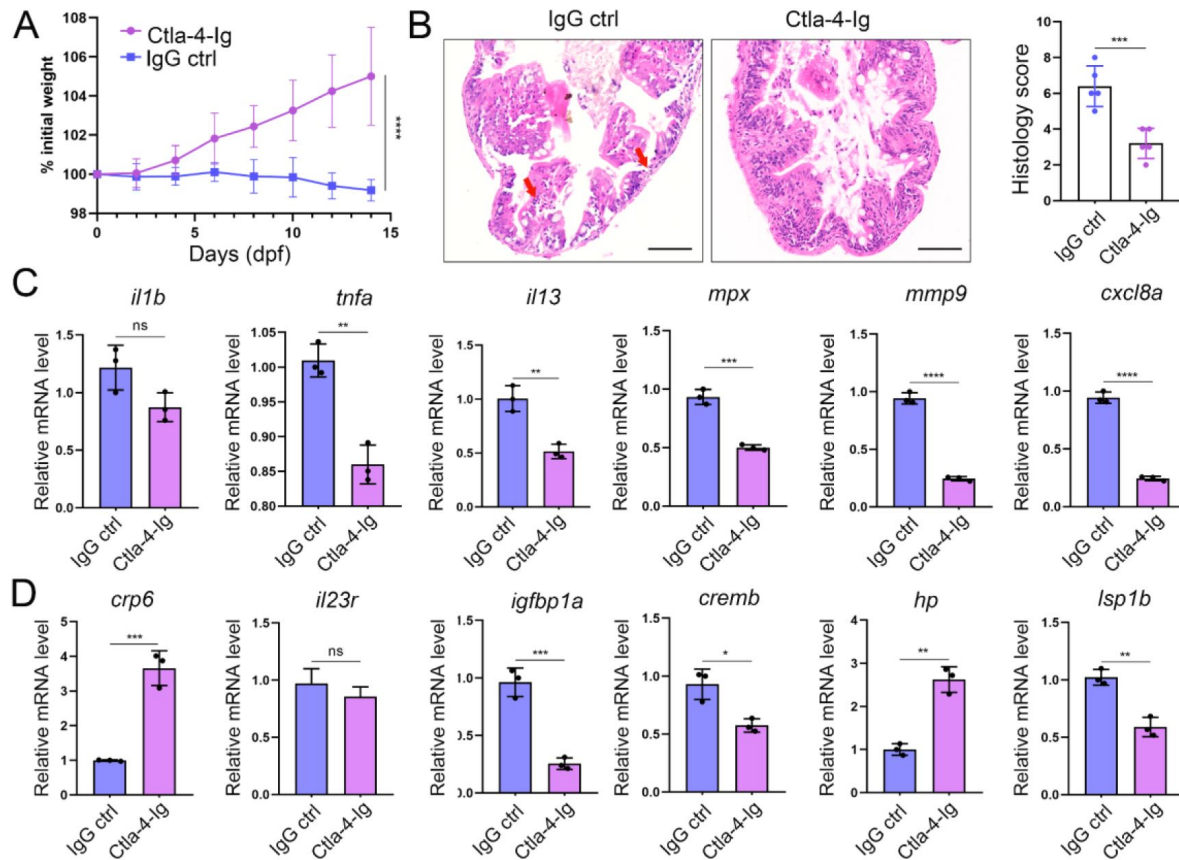


Fig. 8

In vivo inhibition of intestinal inflammation by sCtla-4-Ig.

A Percent initial weight of zebrafish after injection of the sCtla-4-Ig or the IgG isotype control. Each group consisted of six zebrafish (n = 6). Data show means with SEM analyzed by two-way ANOVA with Sidak's correction for multiple comparisons. **B** Representative H&E staining analysis of histopathological changes and quantitation of histology scores in the posterior intestine from *ctla-4*^{-/-} zebrafish treated with sCtla-4-Ig or IgG isotype control. Scale bar: 50 μ m. **C** The mRNA expression levels of inflammation-related genes in *ctla-4*^{-/-} zebrafish treated with sCtla-4-Ig or IgG isotype control. **D** The mRNA expression levels of IBD biomarker genes and IBD-related genes in *ctla-4*^{-/-} zebrafish treated with sCtla-4-Ig or IgG isotype control. The p value was generated by unpaired two-tailed Student's *t*-test. ***p* < 0.01; ****p* < 0.001; *****p* < 0.0001.

despite an increased number of neutrophils and lymphocytes [67]. In summary, Ctla-4 deficiency induces IBD-like phenotypes analogous to those typically elicited by drugs in mice and zebrafish, making this model a valuable tool for comprehending the pathophysiological mechanisms underlying IBD.

A transcriptomics study was conducted to investigate the mechanisms of Ctla-4-deficiency induced IBD. RNA-seq analysis demonstrated a significant upregulation of important inflammatory cytokines, such as *il1b* and *tnfa* in the Ctla-4-deficient intestines. This is consistent with studies showing that IL-1 β and TNF- α act as crucial mediators in mammalian IBD models by disrupting epithelial junctions and inducing apoptosis of epithelial cells [68, 69]. Conversely, the key anti-inflammatory cytokines, such as *il10*, were downregulated. These findings highlight an imbalance between pro-inflammatory and anti-inflammatory cytokines in the intestines of Ctla-4-deficient fish. Consistently, the inflammatory signaling pathways associated with these upregulated cytokines, such as the ERK1/2 and MAPK pathways, were positively enriched in inflamed intestines. Single-cell RNA-seq analysis revealed the upregulation and enrichment of these inflammatory cytokines and pathways in neutrophils, macrophages, and B cells of inflamed intestines, indicating their active involvement in inflammatory responses and as major sources of inflammatory signals. Additionally, there was a marked increase in the abundance of Th2 subset cells in the inflamed intestines, these cells exhibited high expression of *il13* and were significantly enriched in inflammatory signaling pathways, indicating their activated state. These findings align with previous studies indicating that T cells in CTLA-4-deficient mice exhibit a bias toward Th2 differentiation [23, 24]. Furthermore, IL-13, a key effector Th2 cytokine, has been implicated in the pathogenesis of ulcerative colitis in mammals, where it directly contributes to epithelial cell damage by disrupting tight junctions, inducing apoptosis, and impairing cellular restitution [70]. Therefore, upregulated IL13 from Th2 cells may be a significant contributor to the occurrence of intestine inflammation in Ctla-4-deficient zebrafish. Notably, the proportion of ILC3-like cells was downregulated in the inflamed intestines, consistent with recent studies reporting a substantial decline of ILC3 in IBD patients [41–43]. ILC3 is the most abundant type of ILCs in the intestines and plays a protective role in IBD in mammals by promoting epithelial cell proliferation and survival, as well as enhancing intestinal barrier function through the production of IL-22 [44, 46]. Thus, the marked decrease in ILC3-like cells may exacerbate intestinal inflammation and damage.

IBD is frequently associated with alterations in gut microbiota composition, characterized by reduced microbial diversity and an imbalance between beneficial and pathogenic bacteria. The common feature of these changes is the expansion of Proteobacteria, particularly members of the Enterobacteriaceae family [71, 72]. Similarly, Ctla-4-deficient zebrafish exhibited significant enrichment of Enterobacteriaceae, alongside a decline in beneficial bacteria like Cetobacterium and an increase in opportunistic pathogens such as γ -Proteobacteria and Aeromonadales. These findings indicate shared patterns in microbial flora changes during intestinal inflammation. Previous studies suggest that reduced microbial diversity in IBD results from the loss of normal anaerobic bacteria, such as Bacteroides, Eubacterium, and Lactobacillus [73]. Concurrently, inflammation-driven increases in intestinal lumen oxygenation and the availability of nitrate and host-derived electron acceptors facilitate anaerobic respiration and Enterobacteriaceae proliferation [74]. These observations highlight the intricate interplay between IBD pathogenesis, gut microbial alterations, and host immune homeostasis. The zebrafish IBD-like model induced by Ctla-4 deficiency offers new insights into this research area. For instance, abnormal activation of Th2 cells may lead to dysfunction in downstream B cells and mucosa-associated immunity, which are crucial for maintaining symbiotic bacterial homeostasis in the intestines [50]. This suggests a potential link between Th2 cell changes and the observed alterations in the intestinal microbial community in Ctla-4-deficient zebrafish. Moreover, Ctla-4 deficiency alters the proportion and activation of ILC3 cells and damages the intestinal epithelium, potentially shaping the inflammatory milieu and further disrupting gut microbial homeostasis. Ctla-4 also regulates T cell activation by inhibiting the Cd80/86 co-stimulatory pathway. These

findings suggest a regulatory interplay between Ctla-4, ILC3 cells, Cd80/86-primed T cells, and gut microbiota in Ctla-4 deficiency-induced IBD. Recently, gut microbiota exposure has been found to induce local IL-23 production, which upregulates CTLA-4 on ILC3s. This supports immune regulation by reducing CD80/86 co-stimulatory signaling and increasing PD-L1 bioavailability on myeloid cells. Impairment of this pathway manifests in a substantial imbalance of effector and regulatory T cell responses, exacerbating intestinal inflammation [75]. These findings bolster our hypothesis and provide valuable insights into the complex interactions between gut microbiota, ILC3-mediated immune responses, and Cd80/86 signaling in Ctla-4 deficiency-induced IBD.

In conclusion, our study demonstrates that Ctla-4 serves as a potential genetic determinant of the IBD-like phenotype in zebrafish, although further research is necessary to conclusively identify the causative variant responsible for this association. The development of this zebrafish model offers a valuable tool for elucidating the mechanisms underlying the disease's pathophysiology. Nevertheless, a deeper understanding of the intricate interactions among immune cells, intestinal epithelial cells, and the microbiome in IBD remains an area warranting further investigation.

Materials and Methods

Experimental fish

The AB strain zebrafish (*Danio rerio*) of both sexes, 4-6 months of age with body weights ranging from 0.3 to 0.8 g and lengths of 3-4 cm, were reared in the laboratory in recirculating water at 26-28 °C under standard conditions as previously described [76]. All animal experiments were performed in compliance with legal regulations and approved by the Research Ethics Committee of Zhejiang University. For sampling, wild-type and Ctla-4-deficient zebrafish of varying ages were kept in separate tanks and labeled with their respective dates of birth. Wild-type zebrafish aged 4-6 months and Ctla-4-deficient zebrafish aged 4 months were used for the experiments.

Generation of Ctla-4-deficient zebrafish

CRISPR/Cas9 system was used to knock out the *ctla-4* gene. The targeting sequence 5'-CTCAGAGCCCTACTTCGCAA-3' was designed by optimized CRISPR Design (<http://crispr.mit.edu/>) and synthesized by T7 RNA polymerase and purified by MEGAclear Kit (AM1908; Invitrogen) *in vitro*. Cas9 protein (500 ng/μl, A45220P; Thermo Fisher Scientific) and purified RNA (90 ng/μl) were coinjected into one cell-stage wild-type embryos. For genotyping, DNA fragment was amplified with primers (F: 5'-TGTGACAGGAAAAGATGGAGAA-3' and R: 5'-GATCAGATCCACTCCTCCAAAG-3') at 94°C for 4 min followed by 35 cycles at 94°C for 30 s, 58°C for 30 s and 72°C for 30 s, culminating in a final extension at 72°C for 10 min. Subsequently, the PCR product was subjected to sequencing. The mutant alleles (~14 bp) were obtained. As with wild-type zebrafish, Ctla-4-deficient zebrafish were reared in the laboratory in recirculating water at 26-28 °C under standard conditions.

Preparation of recombinant proteins

For prokaryotic expression, the encoding sequences for the extracellular domains of Ctla-4 and Cd80/86 (designated as soluble Cd80/86, sCd80/86) were amplified and cloned into the pET-28a (+) and pCold-GST vectors. The primers used are shown in Supplemental Table 1. The recombinant plasmid was transformed into *Escherichia coli* BL21 (DE3) competent cells (TransGen Biotech) and induced with isopropyl-β-D-thiogalactoside (IPTG, 0.5 mM) at 20°C for 12 h. After ultrasonication, the supernatants were collected for purification. For eukaryotic Ctla-4-Ig and Cd28-Ig expression, the extracellular domains of Ctla-4 and Cd28 were fused to the Fc region of human IgG1 [77], and cloned into the pAcGHLTc vector. The recombinant plasmids were transfected into sf9 (*Spodoptera frugiperda*) cells with baculovirus vector DNA (AB Vector) under the assistance of lipofectamineTM 3000 (Thermo Fisher Scientific). The cells were cultured at 28°C for 3 days and

subsequently dissolved in a lysing buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1% Nonidet P-40, 1 mM PMSF). The recombinant Ctla-4-Ig and Cd28-Ig proteins (designated as soluble Ctla-4-Ig, sCtla-4-Ig and sCd28-Ig) were purified using Ni-NTA agarose affinity chromatography (QIAGEN) following the manufacturer's protocol. Proteins were then separated on a 12% SDS-PAGE gel and visualized through Coomassie Brilliant Blue R250 staining.

Preparation of polyclonal antibody

Antibody (Ab) against the Ctla-4 extracellular domain protein was produced through a recombinant protein immunized approach as previously described [78]. Briefly, four-week-old male BALB/c mice (~15 g) were immunized with recombinant Ctla-4 protein with extracellular domain (25 µg) each time in CFA (Sigma-Aldrich) initially and then in IFA (Sigma-Aldrich) for four times thereafter at biweekly intervals. Seven days after the final immunization, serum samples were collected when anti-serum titers exceeded 1:10,000. The Ab was affinity purified by Protein-A Agarose Columns (Thermo Fisher Scientific), and its titer was examined by ELISA. The validity and specificity of the Ab was determined by Western blot analysis.

Subcellular localization

HEK293T cells were seeded into the 12-well plates (Corning) with cover glass and cultured in high-glucose DMEM (Gibco) supplemented with 10% FBS (Cell-Box) at 37°C in 5% CO₂ to allow growth until 40%-50% confluence. The cells were transfected with pEGFPN1-Ctla-4 plasmid DNA (0.8 µg) with the help of PEI reagent (3.2 µg per well) in accordance with the manufacturer's protocol. After transfection for 24 h, the cells were fixed with 4% paraformaldehyde (PFA; Sigma-Aldrich) and stained with CM-DiI (1 µM; Thermo Fisher Scientific) and DAPI (100 ng/ml; Sigma-Aldrich). The fluorescence images were obtained using a two-photon laser scanning confocal microscope (LSM710; Zeiss, Jena, Germany) with ×630 magnification.

Identification of monomer or dimer

HEK293T cells were transfected with pCDNA3.1-HA-Ctla-4 (0.8 µg) or pCDNA3.1-HA (0.8 µg) under the assistance of polyethylenimine (PEI; Sigma-Aldrich). After 48 h, the cells were lysed with precooled cell lysis buffer (Beyotime) and the supernatants were mixed with non-reducing (without β-ME) or reducing (with β-ME) loading buffer for Western blot analysis.

Immunofluorescence staining

Colocalization of Cd4-1/Cd8α and Ctla-4 was determined by immunofluorescence staining. Leukocytes were isolated from zebrafish's spleen, kidney, and peripheral blood by Ficoll-Hypaque (1.080 g/ml; Sangon Biotech) centrifugation at 2,500 rpm at 25°C for 25 min. After washing with D-Hank's solution, cells were fixed with 4% PFA at room temperature for 10 min, blocked with 2% BSA (Sigma-Aldrich), and incubated with primary Abs at 4°C for 2 h. The primary Abs included combinations of rabbit anti-CD4-1 and mouse anti-CTLA-4, or rabbit anti-CD8α and mouse anti-CTLA-4, which were produced in our laboratory as previously described [75]. Following another wash with D-Hank's solution, the cells were combined with secondary Abs, including FITC-conjugated goat anti-rabbit IgG and PE-conjugated goat anti-mouse IgG (Thermo Fisher Scientific), according to the manufacturer's instructions. After a final wash with D-Hank's solution, the cells were stained with DAPI (100 ng/ml) at room temperature for 10 min. Fluorescence images were captured using a two-photon laser scanning microscope (LSM710; Zeiss, Jena, Germany) with ×630 magnification.

Myeloperoxidase activity measurement

The myeloperoxidase (MPO) activity in intestine and peripheral blood was quantified using a commercial colorimetric assay kit (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's instructions. Briefly, intestinal tissues were homogenized in extraction

buffer to obtain a 5% homogenate, while peripheral blood was mixed with extraction buffer at a 1:1 ratio. A 180 μ l aliquot of resultant mixture was incubated with 20 μ l of reaction buffer for 15 min at 37°C. Enzymatic activity was determined by measuring the changes in absorbance at 460 nm using a 96-well plate reader. MPO activity was expressed as units per gram of wet intestinal tissue or per milliliter of peripheral blood.

Co-immunoprecipitation and Western blot analysis

Co-immunoprecipitation (Co-IP) was performed to detect the interaction between Cd28/Ctla-4 and Cd80/86. HEK293T cells were co-transfected with pLVX-mCherry-C1-Cd28 (3 μ g) and pEGFP-N1-Cd80/86 (3 μ g) or pEGFP-N1-Ctla-4 (3 μ g) and pCDNA3.1-HA-Cd80/86 (3 μ g) using PEI as a transfection reagent. At 48 h post-transfection, the cells were lysed with pre-cooled cell lysis buffer (Beyotime). The lysates were centrifuged at $12,000 \times g$ for 8 minutes at 4°C, and the supernatants were incubated with mouse anti-myc mAb (Abmart) or mouse anti-HA mAb (Abmart) overnight at 4°C. Expression of the transfected plasmids was analyzed in the whole cell lysates as an input control. The following day, the mixture was incubated with 50 μ l of protein A agarose beads (Thermo Fisher Scientific) for 4 h. The beads were washed 3 times with lysis buffer and mixed with loading buffer for SDS-PAGE separation. Target proteins were transferred onto a 0.22- μ m polyvinylidene difluoride (PVDF) membrane (EMD Millipore) for Western blot analysis. The blots were blocked with 5% skimmed milk, incubated with mouse or rabbit primary Abs overnight at 4°C, washed with TBST, and then incubated with HRP-conjugated goat anti-mouse/rabbit IgG mAb (Abmart) at room temperature for 1 h. Detection was performed using a gel imaging system (Tanon 4500).

Histopathological analysis

The anterior, mid, and posterior intestines (n = 3 replicates) were fixed in 4% PFA overnight and embedded in paraffin. The tissues were cut into 4 μ m sections and stained with hematoxylin and eosin (H&E) for histopathological analysis. To evaluate the severity of intestinal inflammation, histologic scores were determined based on established criteria from a previous study [79]. Briefly, three independent parameters, including inflammation severity, inflammation extent, and epithelial changes, were assessed and scored as follows: inflammation severity (0 = none, 1 = minimal, 2 = mild, 3 = moderate, 4 = marked); inflammation extent (0 = none, 1 = mucosa, 2 = mucosa and submucosa, 3 = transmural), epithelial changes (0 = none, 1 = minimal hyperplasia, 2 = mild hyperplasia, minimal goblet cell loss, 3 = moderate hyperplasia, mild goblet cell loss, 4 = marked hyperplasia with moderate to marked goblet cell loss). Each parameter was calculated and summed to obtain the overall score. Additionally, tissue sections were stained with Periodic Acid-Schiff (PAS) or Alcian Blue and Periodic Acid-Schiff (AB-PAS) reagent to evaluate the mucin components and goblet cell numbers.

Transmission electron microscopy observation

The posterior intestines were cut into 0.2 cm segments and then split lengthwise to expose the intestinal villi to the fixative fully. The samples were first fixed with 2.5% glutaraldehyde in phosphate buffer (0.1 M, pH 7.0) overnight at 4°C, washed three times in the phosphate buffer for 15 min at each step and post-fixed in 1% OsO₄ for 1 h. Following gradient acetone dehydration and Spurr resin infiltration (1:1 and 1:3 mixture of absolute acetone and the final Spurr resin mixture for 1 h and 3 h, and final Spurr resin mixture overnight), the specimens were placed in Eppendorf contained Spurr resin and heated at 70°C for overnight. The samples were sectioned using an ultratome (LEICA EM UC7). Then, the sections were stained with uranyl acetate and alkaline lead citrate for 10 min each and observed under a transmission electron microscopy (Hitachi Model H-7650).

Assessment of apoptosis by TUNEL

The posterior intestines were fixed by 4% paraformaldehyde and embedded in paraffin. Apoptosis was detected using TUNEL staining following the manufacturer's protocol (Beyotime). Briefly, deparaffinized sections were incubated with biotin-dUTP labeling solution (TdT Enzyme: Biotin-dUTP = 1: 9) for 1 h, followed by incubation with streptavidin-HRP for 30 min at room temperature. Positive signals were visualized using DAB chromogenic solution and counterstained with hematoxylin. Apoptotic cells and the area of the intestinal epithelium were quantified, and the apoptosis index was calculated as the number of apoptotic cells per $1 \times 10^4 \mu\text{m}^2$ observed using ImageJ software.

RNA-sequencing and bioinformatic analysis

Total RNAs were isolated from wild-type or *ctla-4*^{-/-} intestines (three biological replicates) using TRIzol reagent following the manufacturer's instructions (Takara). cDNA libraries were constructed using NEB Next Ultra Directional RNA Library Prep Kit (NEB), and sequencing was performed according to the Illumina HiSeq2500 standard protocol at LC Bio (Hangzhou, China). The differentially expressed genes (DEGs) were identified with absolute Log₂ fold change > 1 and adjusted *p*-value < 0.05 by R package DESeq2. Gene Ontology (GO) enrichment and Kyoto encyclopedia of genes and genomes (KEGG) enrichment analyses were performed by the OmicStudio (<http://www.omicstudio.cn/tool>) tools. Gene-set enrichment analysis was performed using software GSEA (v4.1.0, <https://www.gsea-msigdb.org/gsea/index.jsp>), and the heatmap was generated using the R package ggplot2. For the protein-protein interaction (PPI) networks, the DEGs were retrieved in STRING (version 11.5, <https://string-db.org/>) database (combined score > 0.4), and the PPI network was visualized by Cytoscape software (version 3.9.1) [80]. The betweenness centrality (BC) was calculated using the CytoNCA plugin in Cytoscape software. The RNA-sequencing (RNA-seq) data in this study were deposited in the Gene Expression Omnibus (GEO) (<http://www.ncbi.nlm.nih.gov/geo/>) database.

Quantitative real-time PCR

The transcript abundance of target genes was quantified using quantitative real-time PCR on a CFX Connect Real-Time PCR Detection System equipped with Precision Melt Analysis™ Software (Bio-Rad, Cat. No. 1855200EM1). Total RNA from intestines was extracted using TRIzol reagent (Takara Bio) and reverse transcribed into cDNAs according to the manufacturer's protocol. PCR experiments were performed in a total volume of 10 μl by using an SYBR Premix Ex Taq kit (Takara Bio). The reaction mixtures were incubated for 2 min at 95°C, then subjected to 40 cycles of 15 s at 95°C, 15 s at 60°C, and 20 s at 72°C. Relative expression levels of the target genes were calculated using the $2^{-\Delta\Delta\text{Ct}}$ method with β -actin for normalization. Each PCR trial was run in triplicate parallel reactions and repeated three times. The primers used in the experiments are listed in Supplemental Table 1.

Single-cell RNA-sequencing analysis

The intestines from wild-type (30 fish) and *ctla-4*^{-/-} zebrafish (30 fish) were washed by D-Hank's and incubated with D-Hank's containing EDTA (0.37 mg/mL) and DTT (0.14 mg/mL) for 20 min. The resulting supernatants were collected as fraction 1. The remaining tissues were subsequently digested with type IV collagenase (0.15 mg/mL) for 1 h at room temperature and the resulting supernatants were collected as fraction 2. Fractions 1 and 2 were combined and centrifuged at 350 g for 10 min. Cells were then washed with D-Hank's and suspended in a 40% Percoll (Shanghai Yes Service Biotech, China) solution (Percoll: FBS: L-15 medium = 4: 1: 5) and passed through a 40- μm strainer (Bioland). The cell suspension was carefully layered onto 63% Percoll (Percoll: FBS: L-15 medium = 6.3: 1: 2.7) and centrifuged at 400 g for 30 min at room temperature. The cell layer at the interface was collected and washed with D-Hank's at 400 g for 10 min. Cell quantity and viability were assessed using 0.4% trypan blue staining, revealing that over 90% of the cells were viable.

Single-cell samples (8,047 cells in wild-type group, 8,321 cells in *ctla-4*^{-/-} group) were submitted to the LC-Bio Technology Co., Ltd (Hangzhou, China) for 10× Genomics library preparation and data analysis assistance. Libraries were prepared using the Chromium™ Controller and Chromium™ Single Cell 3' Library & Gel Bead Kit v2 (10× Genomics) according to the manufacturer's protocol, and sequenced on an Illumina NovaSeq 6000 sequencing system (paired-end multiplexing run, 150 bp) at a minimum depth of 20,000 reads per cell. Sequencing results were demultiplexed and converted to FASTQ format using Illumina bcl2fastq software and the data were aligned to Ensembl zebrafish genome assembly GRCz11. Quality control was performed using the Seurat. DoubletFinder R package was used to identify and filter the doublets (multiplets) [81]. The cells were removed if they expressed fewer than 500 unique genes, or > 60% mitochondrial reads. Cloud-based Cumulus v1.0 was used to perform the batch correction (using the Harmony algorithm) on the aggregated gene-count matrices [82]. The number of cells after filtration in the current study was 3,263 in wild-type and 4,276 in *ctla-4*^{-/-} groups, respectively. Cells were grouped into an optimal number of clusters for *de novo* cell type discovery using Seurat's FindNeighbors() and FindClusters() functions, graph-based clustering approach with visualization of cells being achieved through the use of t-SNE or UMAP plots [83]. The cell types were determined using a combination of marker genes identified from the literature and gene ontology. The marker genes were visualized by dot plot and t-SNE plots to reveal specific individual cell types.

16S rRNA sequencing analysis

Intestinal contents were collected from both wild-type and *ctla-4*^{-/-} zebrafish by gently squeezing the intestines with fine-tipped tweezers, and the remaining intestines were used for single-cell RNA-sequencing analysis. Contents from six fish were pooled to form one replicate, with each experimental sample comprising four replicates. DNA was extracted from the samples using the CTAB method, a protocol known for its efficacy in isolating DNA from trace quantities. The quality of DNA was assessed through agarose gel electrophoresis. Total DNA was amplified to construct sequencing libraries using primers (341F: 5'-CCTACGGGNGGCWGCAG-3'; 805R: 5'-GACTACHVGGGTATCTAATCC-3') targeting the V3-V4 regions of the 16S rRNA gene. The libraries were sequenced on an Illumina NovaSeq PE250 platform. Quality filtering was performed under specific conditions to obtain high-quality clean tags using fqtrim (v0.94). Chimeric sequences were removed using Vsearch software (v2.3.4). After dereplication using DADA2, a feature table and feature sequences were generated. Alpha diversity is applied in analyzing the complexity of species diversity for a sample through the Shannon and Simpson indices, with all calculations performed in QIIME2. Beta diversity analysis was also conducted in QIIME2, and the graphs were drawn by R package. Sequence alignment was performed using BLAST, and representative sequences were annotated using the SILVA database. Other diagrams were implemented using the R package (v3.5.2).

In vitro lymphocyte proliferation assay

The leukocytes were prepared from the spleen, kidney, and peripheral blood of wild-type (10 fish) or *ctla-4*^{-/-} (10 fish) zebrafish through Ficoll-Hypaque centrifugation. A total of 2×10^6 leukocytes from either wild-type or *ctla-4*^{-/-} individuals were labeled with 1 μM carboxyfluorescein succinimidyl ester (CFSE; Thermo Fisher Scientific) at 25°C for 5 min. The reaction was terminated by adding a triploid volume of Leibovitz L-15 medium (Gibco) supplemented with 10% FBS, as previously described [84]. After washing with D-Hank's solution, the cells were cultured in L-15 medium containing 10% FBS in the presence or absence of PHA (5 μg/ml), recombinant Ctla-4-Ig (20 μg/ml), Cd28-Ig (20 μg/ml), Cd80/86 (10 μg/ml) proteins or anti-Ctla-4 Ab (10 μg/ml) at 28°C for 3 days. CFSE fluorescence intensity in the labeled co-cultures was analyzed using a flow cytometer (FACSJazz; BD Biosciences) to assess cell division [85].

Prediction of protein interactions by AlphaFold2

AlphaFold2 (version 2.3.2; available at <https://github.com/google-deepmind/alphafold>) was implemented on a high-performance computing cluster to predict the structures of the Cd80/86 complexes with Cd28 and Ctla-4 [85]. The resulting models were ranked based on their per-residue Local Distance Difference Test (pLDDT) scores, which quantify the confidence level of each residue on a scale from 0 to 100. Residues were color-coded according to their pLDDT scores, with higher values reflecting greater confidence in the prediction. Furthermore, the Predicted Aligned Error (PAE) scores were analyzed to identify well-defined interaction interfaces between Cd28 or Ctla-4 and Cd80/86.

Microscale thermophoresis assay

The binding affinity between Cd80/86 and Cd28/Ctla-4 were measured through microscale thermophoresis (MST) assays using a Monolith NT.115 instrument (Nano Temper Technologies) as previously described [86]. In each assay, the labeled proteins (Cd28/Ctla-4 with EGFP-tag) were incubated with varying concentrations of unlabeled ligand protein (Cd80/86) for 10 min at room temperature. The initial protein concentration of 3.2 μ M was diluted into 16 different concentrations by doubling dilution. The samples were then loaded into capillaries and analyzed at 25°C by using 40% light emitting diode (LED) and medium MST power. The binding affinities of Cd80/86 with Cd28 and Cd80/86 with Ctla-4 were examined using the same parameters. Each assay was repeated three to five times, and dissociation constants (K_D) were calculated using MO.Affinity Analysis software.

In vivo administration of sCtla-4-Ig

An *in vivo* sCtla-4-Ig administration assay was conducted to evaluate the potential therapeutic effect of sCTLA-4-Ig on intervention of a *ctla-4*-deficiency induced IBD-like phenotype. For this procedure, the *ctla-4*^{-/-} zebrafish were i.p administered with recombinant soluble Ctla-4-Ig protein (sCtla-4-Ig, 2 μ g/g body weight) every other day until day 14. Fish that received an equal amount of human IgG isotype were used as control. The dose of sCtla-4-Ig was chosen based on its ability to significantly inhibit the mRNA expression of *il13* in Ctla-4-deficient zebrafish.

Statistical analysis

Statistical analysis and graphical presentation were performed with GraphPad Prism 8.0. All data were presented as the mean $\pm$ SD of each group. Statistical evaluation of differences was assessed using one-way ANOVA, followed by an unpaired two-tailed *t*-test. Statistical significance was defined as **p* < 0.05, ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001. All experiments were replicated at least three times.

Supplemental Figures

Fig. S2

Preparation of mouse anti-Ctla-4 antibody.

A SDS-PAGE detection of the recombinant Ctla-4 protein with extracellular domain (ECD). Lane 1, 2 and 3 represent the protein markers, blank, and target protein, respectively. **B** Western blot analysis of the mouse anti-EGFP and anti-Ctla-4 antibodies that bind to the recombinant Ctla-4-EGFP fusion proteins expressed in HEK293T cells. **C** Western blot analysis of native Ctla-4 protein in zebrafish intestinal tissues using mouse anti-Ctla-4 antibody.

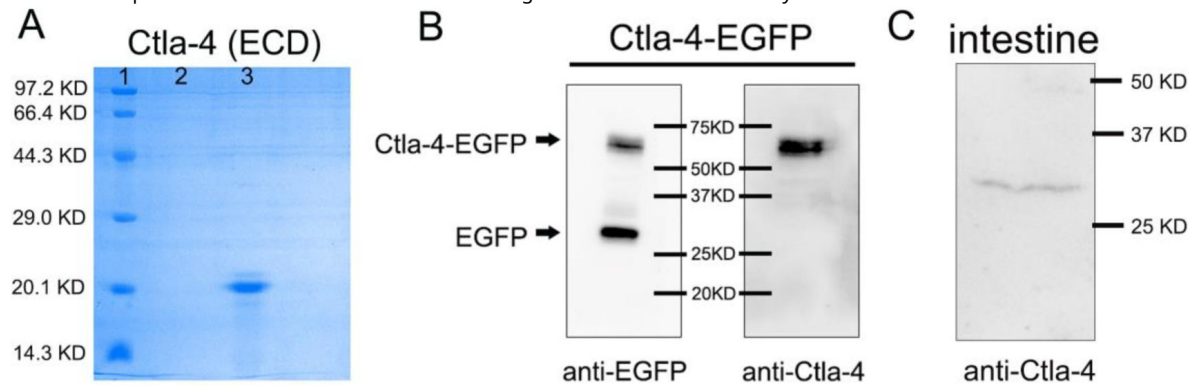
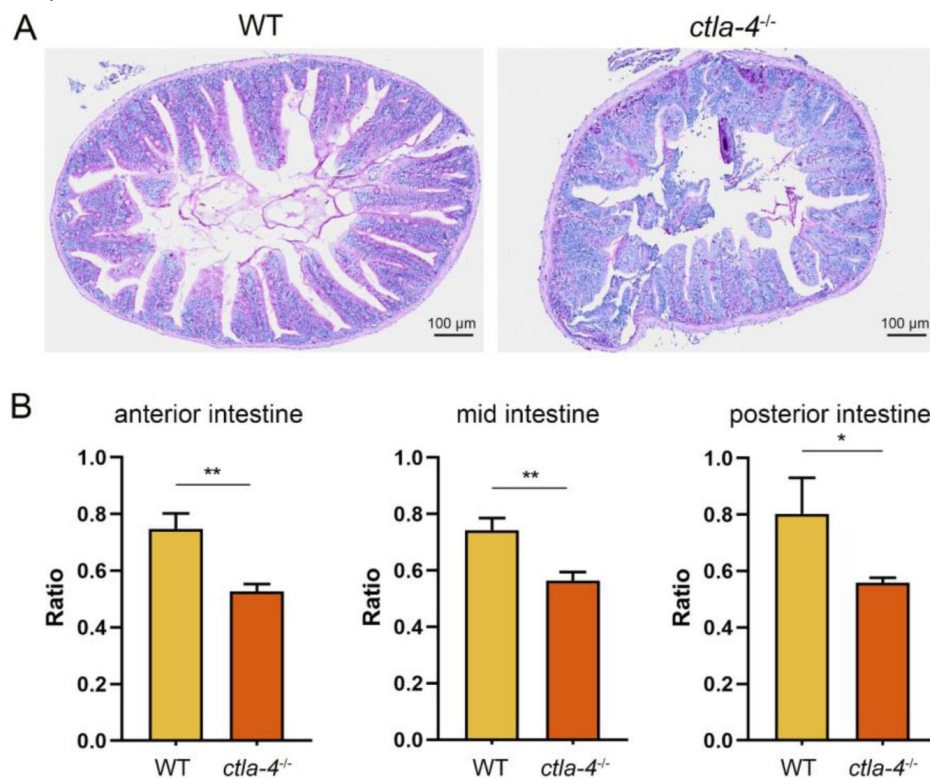


Fig. S3

Histopathological analysis of intestines.

A Periodic Acid-Schiff (PAS) staining was used to analyze the mucin components in anterior intestine from wild-type (WT) and *ctla-4*^{-/-} zebrafish (n = 5). **B** The ratio of intestinal villi length to intestinal ring radius was measured in the anterior, mid, and posterior intestines of WT and *ctla-4*^{-/-} zebrafish (n = 6). Statistical significance was assessed through an unpaired Student's t test (**p* < 0.05; ***p* < 0.01).



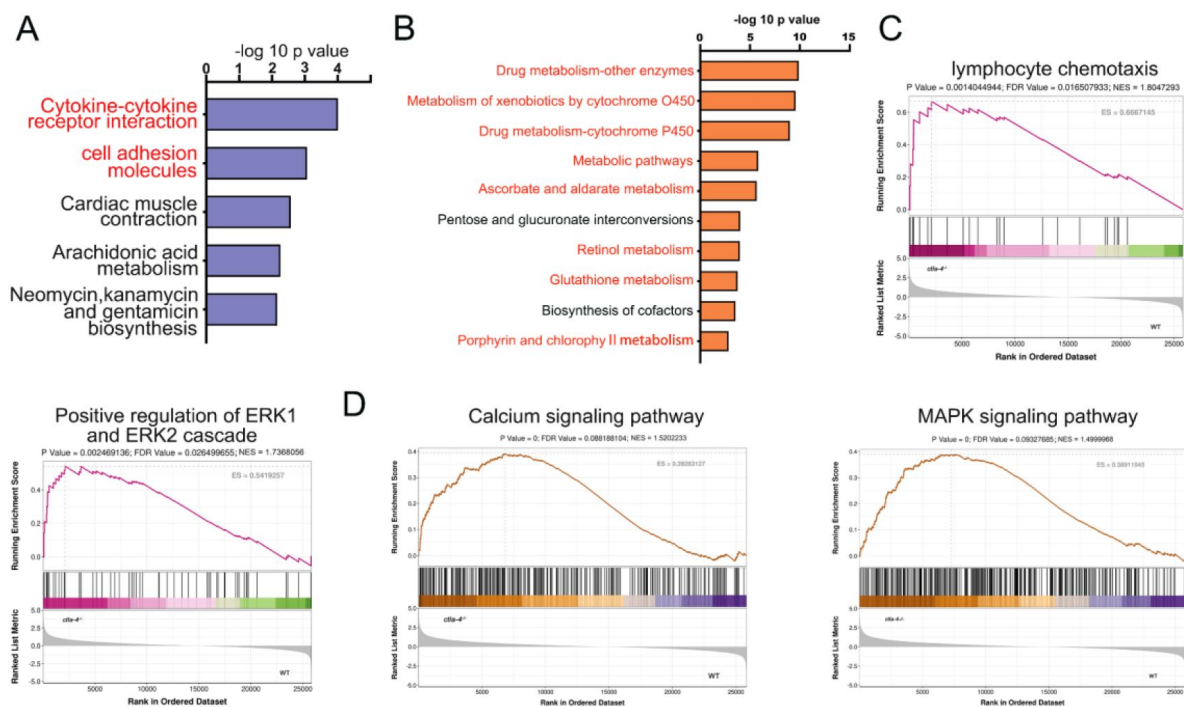


Fig. S4

Examination on the functional genes and pathways associated with the IBD-like phenotype in *ctla-4*^{-/-} zebrafish.

A Top 5 KEGG enrichment bar plot of up-regulated genes in *ctla-4*^{-/-} zebrafish intestines versus wild-type (WT) zebrafish intestines. **B** Top 10 KEGG enrichment bar plot of down-regulated genes in *ctla-4*^{-/-} zebrafish intestines versus WT zebrafish intestines. **C, D** Changes in the expression of genes associated with lymphocyte chemotaxis, positive regulation of ERK1/ERK2 cascades, Calcium and MAPK signaling pathways in the *ctla-4*^{-/-} zebrafish intestines analyzed by using a collection of pre-defined gene sets retrieved from GO (**C**) and KEGG (**D**) database. The *p* value, false discovery rates (FDR) and normalized enrichment score (NES) are shown above each pathway graph.

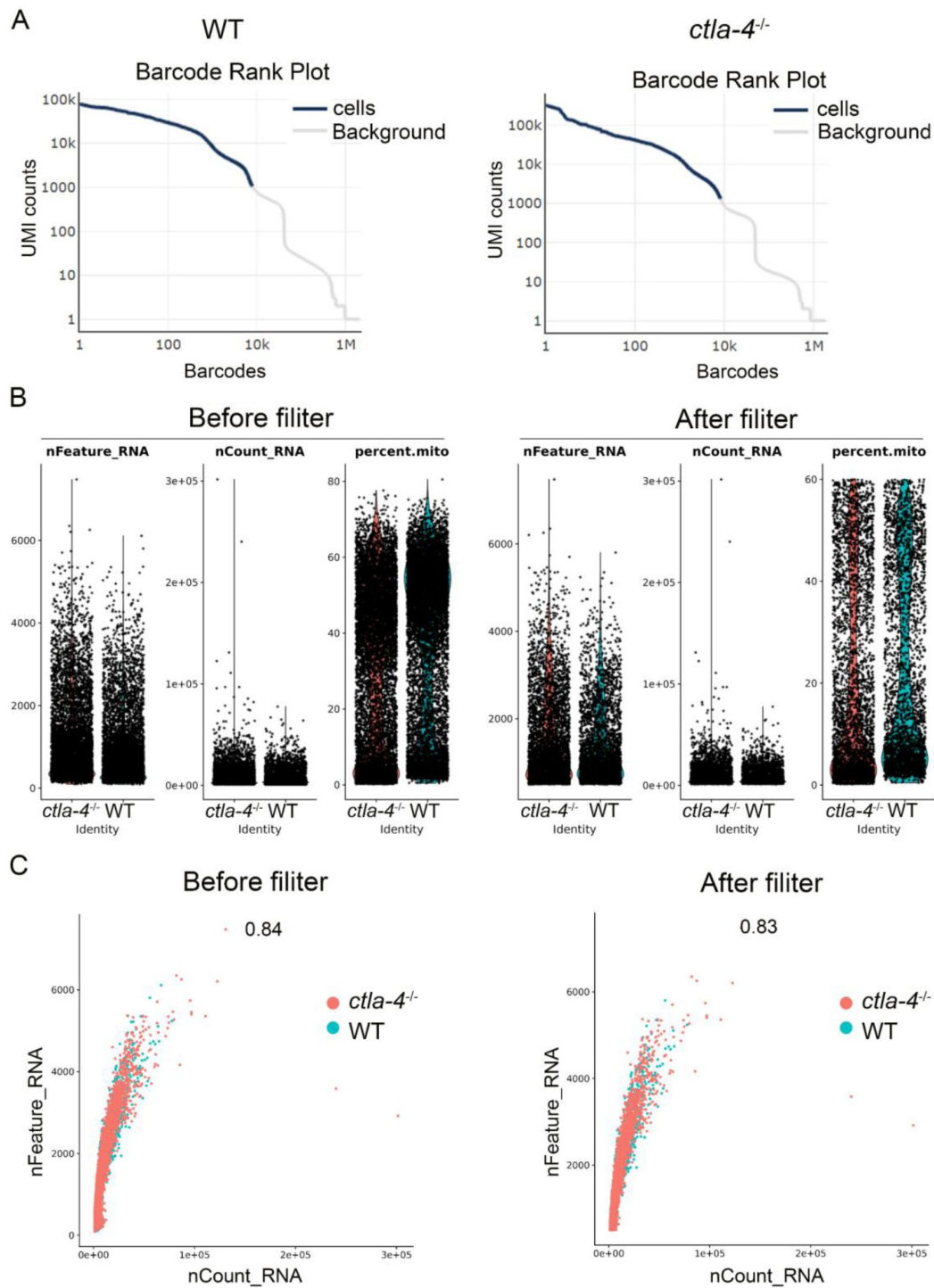


Fig. S5

Quality control analysis of single-cell RNA sequencing data.

A Identification of the effective cell number of the sample. The blue line represents the effective cells corresponding to barcodes, while the gray line denotes the background noise. **B** The basic cellular metrics before and after filtering, including the total number of detected genes (nFeature_RNA), the total number of UMIs (nCount_RNA), and the percentage of reads mapped to mitochondrial genes (Percent.mito). **C** The scatter plot comparing the cellular metrics before and after filtering, showing the relationship between nCount_RNA and nFeature_RNA. The Pearson correlation coefficients are indicated above the graph.

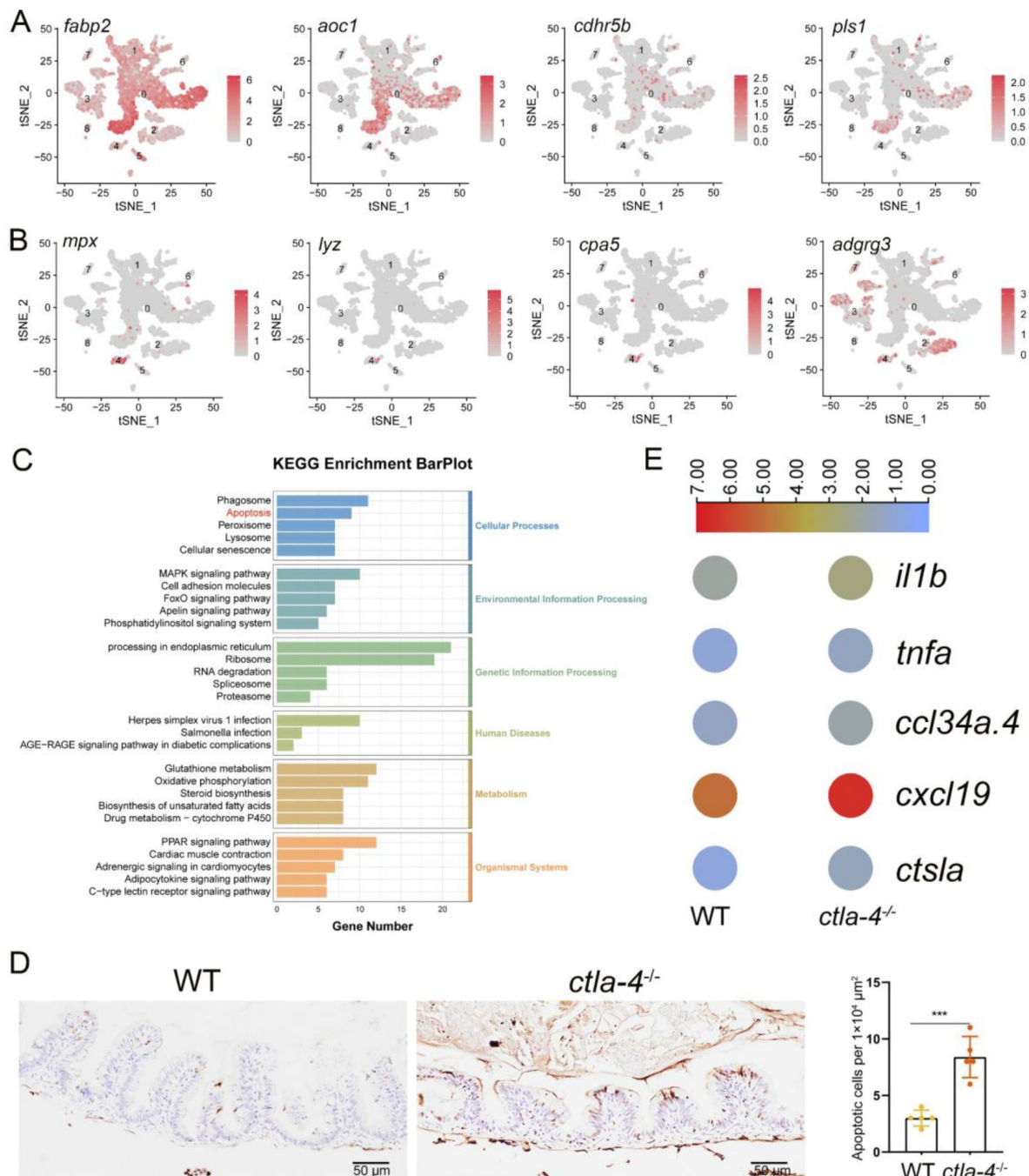


Fig. S6

Examination on the involvement of apoptotic process in epithelial cells and expression of inflammation-related genes in neutrophils and B cells in the intestines of *ctla-4^{-/-}* zebrafish.

A Expression map of the epithelial markers within the cell populations of the zebrafish intestines. **B** Expression map of the neutrophil markers within the cell populations of the zebrafish intestines. **C** KEGG enrichment bar plot of all differentially expressed genes (DEGs) from epithelial cells. **D** Quantification of TUNEL-positive cells per $1 \times 10^4 \mu\text{m}^2$ in WT and *ctla-4^{-/-}* posterior intestines (n = 5). **E** Heatmap of inflammation-related genes in B cells from wild-type and *ctla-4^{-/-}* intestines. Statistical significance was assessed through an unpaired Student's t test ($***p < 0.001$).

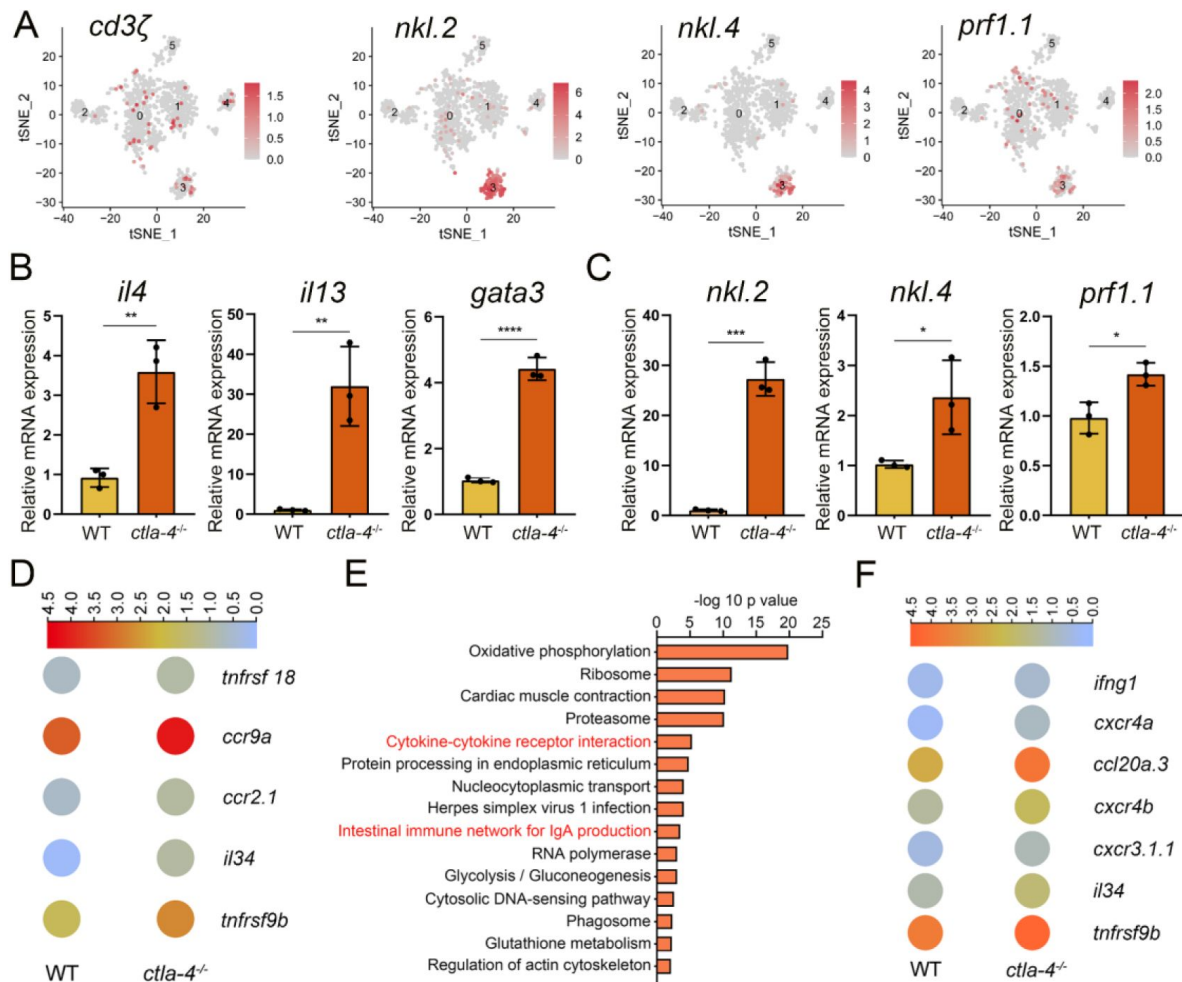


Fig. S7

Examination on the activation of T cell subsets in the intestines of *ctla-4^{-/-}* zebrafish.

A Marker gene expression in individual cluster identifies the cluster as NKT-like cells. **B** RT-qPCR confirms the mRNA expression levels of Th2 cell marker genes in the intestines of wild-type (WT) and *ctla-4^{-/-}* zebrafish. **C** RT-qPCR validates the mRNA expression levels of NKT-like cell marker genes in the intestines of WT and *ctla-4^{-/-}* zebrafish. **D** Heatmap illustrates up-regulated genes involved in cytokine-cytokine receptor interaction in NKT-like cells from WT and *ctla-4^{-/-}* samples. **E** KEGG enrichment analysis reveals the top 15 terms of up-regulated genes in Cd8⁺ T cells in *ctla-4^{-/-}* samples versus WT samples. **F** Heatmap displays up-regulated genes involved in cytokine-cytokine receptor interaction in Cd8⁺ T cells from WT and *ctla-4^{-/-}* samples. Statistical significance was assessed through an unpaired Student's t test (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$).

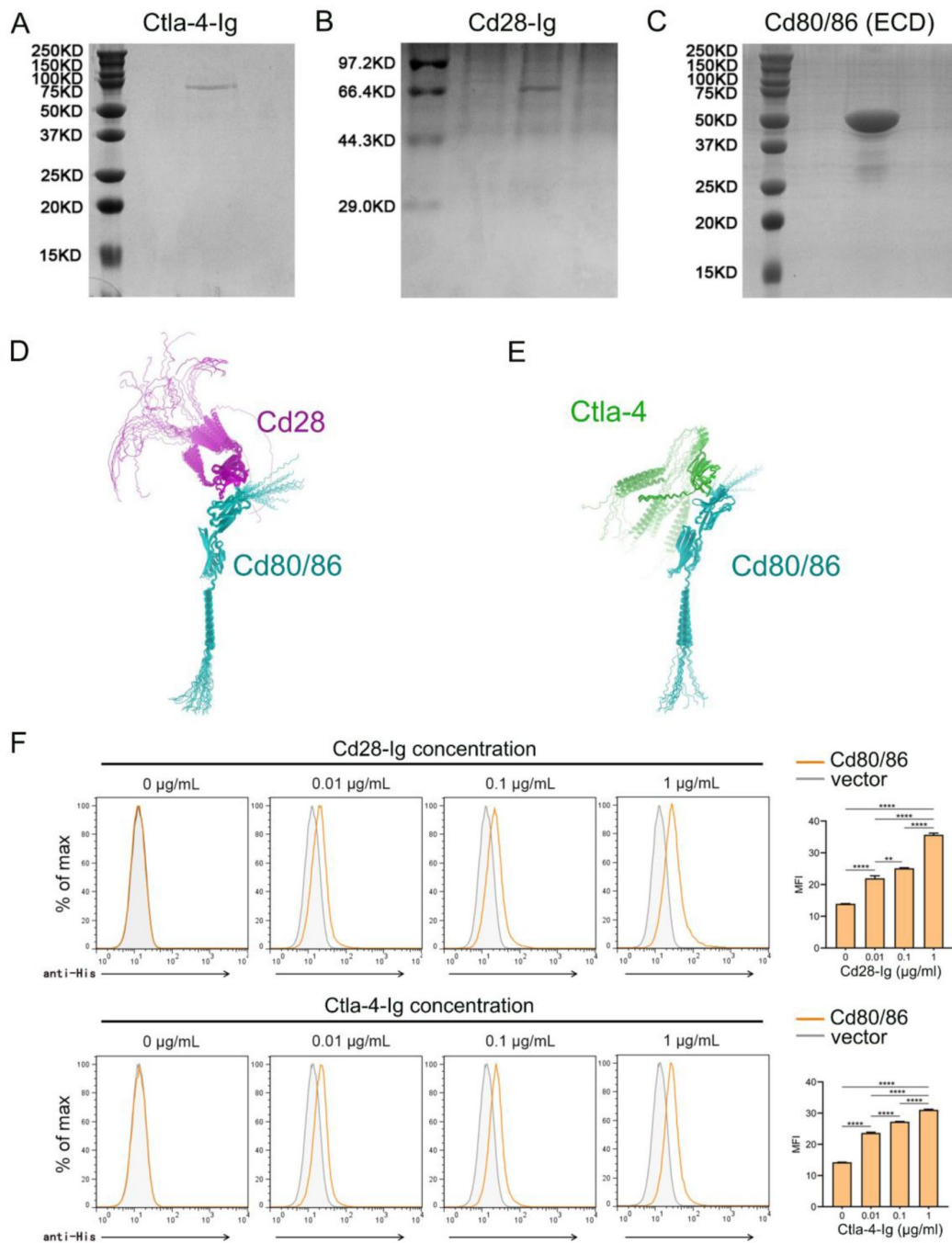


Fig. S8

Preparation of recombinant proteins and examination of their molecular interactions.

A-C SDS-PAGE detection of the purified recombinant soluble Ctla-4-Ig (sCtla-4) (**A**) and sCd28-Ig (**B**) proteins and the Cd80/86 extracellular domain (ECD) (**C**) with Coomassie brilliant blue staining. **D-E** The predicted molecular interactions between Cd80/86 and Cd28 (**D**), as well as Cd80/86 and Ctla-4 (**E**), as modeled by AlphaFold2. The structures are represented in a cartoon style, with Cd80/86, Cd28, and Ctla-4 colored cyan, green, and magenta, respectively. A total of 25 models were predicted for each complex and aligned with Cd80/86. **F** Flow cytometry analysis of the interactions between Cd80/86 and Cd28 (top), and Cd80/86 and Ctla-4 (bottom). Cd80/86 was expressed on HEK293T cells and incubated with varying concentrations of fluorescently labeled sCd28-Ig or sCtla-4-Ig. Fluorescence intensity was detected by flow cytometry to determine molecular interactions. Data are presented as mean $\pm$ SD, derived from three independent experiments. Statistical significance was evaluated using an unpaired Student's *t* test (** $p < 0.01$; *** $p < 0.001$).

Data availability

All data generated or analyzed during this study are included in this article and its supplementary information files. The RNA-seq and scRNA-seq data for this study have been deposited in NCBI Gene Expression Omnibus (GEO) (<https://www.ncbi.nlm.nih.gov/geo/>) under accession numbers GSE255304 and GSE255303, respectively. The 16S rRNA sequencing data in this study have been deposited in the NCBI Sequence Read Archive (SRA) (<https://www.ncbi.nlm.nih.gov/sra/>) with an accession number of BioProject PRJNA1073727.

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Additional information

Author Contributions

L. L. Q. and J. Z. S. conceived and designed the experiments. L. L. Q. and C. B. H. conducted the experiments. Data analysis was performed by L. L. Q., C. B. H., Q. Z., Y. W., A. F. L., L. X. X., Y. C., and J. Z. S. Reagents, materials, and analysis tools were contributed by L. X. X., D. D. F., and J. Z. S. The manuscript was written by L. L. Q., Y. C., L. X. X., and J. Z. S. All authors reviewed the manuscript and provided feedback.

Ethical approval and consent to participate

All animal experiments were performed with the approval of the Ethics Committee for Animal Experimentation of Zhejiang University.

Additional files

Primers used in the study [↗](#)

The DEGs of *ctla-4*^{-/-} vs WT (RNA-seq analysis) [↗](#)

scRNA-seq analysis-1 [↗](#)

scRNA-seq analysis-2 [↗](#)

Animal Welfare and Ethics [↗](#)

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Reviewer #1 (Public review):

"Unraveling the Role of Ctla-4 in Intestinal Immune Homeostasis: Insights from a novel Zebrafish Model of Inflammatory Bowel Disease" generates a 14bp deletion/early stop codon mutation that is viable in a zebrafish homolog of ctla-4. This mutant exhibits an IBD-like phenotype, including decreased intestinal length, abnormal intestinal folds, decreased goblet cells, abnormal cell junctions between epithelial cells, increased inflammation, and alterations in microbial diversity. Bulk and single-cell RNA-seq show upregulation of immune and inflammatory response genes in this mutant (especially in neutrophils, B cells, and macrophages) and downregulation of genes involved in adhesion and tight junctions in mutant enterocytes. The work suggests that the makeup of immune cells within the intestine is altered in these mutants, potentially due to changes in lymphocyte proliferation. Introduction of recombinant soluble Ctla-4-Ig to mutant zebrafish rescued body weight, histological phenotypes, and gene expression of several pro-inflammatory genes, suggesting a potential future therapeutic route.

Strengths:

- Generation of a useful new mutant in zebrafish *ctla-4*
- The demonstration of an IBD-like phenotype in this mutant is extremely comprehensive.
- Demonstrated gene expression differences provide mechanistic insight into how this mutation leads to IBD-like symptoms.
- Demonstration of rescue with a soluble protein suggests exciting future therapeutic potential
- The manuscript is mostly well organized and well written.

Initial Weaknesses were addressed during review.

<https://doi.org/10.7554/eLife.101932.2.sa3>

Reviewer #2 (Public review):

Summary:

The authors aimed to elucidate the role of *Ctla-4* in maintaining intestinal immune homeostasis by using a novel *Ctla-4*-deficient zebrafish model. This study addresses the challenge of linking CTLA-4 to inflammatory bowel disease (IBD) due to the early lethality of CTLA-4 knockout mice. Four lines of evidence were shown to show that *Ctla-4*-deficient zebrafish exhibited hallmarks of IBD in mammals: 1) impaired epithelial integrity and infiltration of inflammatory cells; 2) enrichment of inflammation-related pathways and the imbalance between pro- and anti-inflammatory cytokines; 3) abnormal composition of immune cell populations; and 4) reduced diversity and altered microbiota composition. By employing various molecular and cellular analyses, the authors established *ctla-4*-deficient zebrafish as a convincing model of human IBD.

Strengths:

The characterization of the mutant phenotype is very thorough, from anatomical to histological and molecular levels. The finding effectively established *ctla-4* mutants as a novel zebrafish model for investigating human IBD. Evidence from the histopathological and transcriptome analysis was very strong and supports a severe interruption of immune system homeostasis in the zebrafish intestine. Additional characterization using s*Ctla-4*-Ig further probed the molecular mechanism of the inflammatory response, and provided a potential treatment plan for targeting *Ctla-4* in IBD models.

Weaknesses:

To probe the molecular mechanism of *Ctla-4*, the authors used a spectrum of antibodies that target *Ctla-4* or its receptors. The phenotype assayed was lymphocyte proliferation, while it was the composition rather than number of immune cells that was observed to be different in the scRNASeq assay. Although s*Ctla-4* has an effect of alleviating the IBD-like phenotypes, I found this explanation a bit oversimplified.

Comments on revised version:

The authors have sufficiently addressed all my concerns and I don't have further suggestions.

<https://doi.org/10.7554/eLife.101932.2.sa2>

Reviewer #3 (Public review):

Summary:

Current study on the mutant zebrafish for IBD modeling is worth trying. The author provided lots of evidence, including histopathological observation, gut microflora, as well as intestinal tissue or mucosa cells' transcriptomic data. The multi-omic study has demonstrated the enteritis pathology at multi levels in zebrafish model.

Strengths:

The important immune checkpoint of Treg cells were knockout in zebrafish, and the enteritis were found then. It could be a substitution of mouse knockout model to investigate the molecular mechanism of gut disease.

Weaknesses:

- (1) In Fig. 2I, as to the purple glycogen signals stained by PAS was ignored for the quantitative statistics. The purple stained area could be calculated by ImageJ.
- (2) Those characters in Fig. 3G are too small to recognize. It is suggested to adjusted this picture or just put it in the supplementation, with bigger size.
- (3) The tissue seems damaged for IgG ctrl in Fig. 8B. It is suggested to find another slice to present here.
- (4) Line 667 & 743: "16S rRNA sequencing" should be "16S rRNA gene sequencing". Please check this point throughout the text.

<https://doi.org/10.7554/eLife.101932.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1:

The manuscript suggests the zebrafish homolog of ctla-4 and generates a new mutant in it. However, the locus that is mutated is confusingly annotated as both CD28 (current main annotation in ZFIN) and CTLA-4/CD152 (one publication from 2020), see: <https://zfin.org/ZDB-GENE-070912-128>. Both human CTLA-4 and CD28 align with relatively similar scores to this gene. There seem to be other orthologs of these receptors in the zebrafish genome, including CD28-like (<https://zfin.org/ZDB-GENE-070912-309>) which neighbors the gene annotated as CD28 (exhibiting similar synteny as human CD28 and CTLA-4). It would be helpful to provide more information to distinguish between this family of genes and to further strengthen the evidence that this mutant is in ctla-4, not cd28. Also, is one of these genes in the zebrafish genome (e.g. cd28l) potentially a second homolog of CTLA-4? Is this why this mutant is viable in zebrafish and not mammals? Some suggestions:

(a) A more extensive sequence alignment that considers both CTLA-4 and CD28, potentially identifying the best homolog of each human gene, especially taking into account any regions that are known to produce the functional differences between these receptors in mammals and effectively assigns identities to the two genes annotated as "cd28" and "cd28l" as well as the gene "si:dkey-1H24.6" that your CD28 ORF primers seem to bind to in zebrafish.

In response to the reviewer's insightful suggestions, we have conducted more extensive sequence alignment and phylogenetic analyses that consider both CTLA-4, CD28, and CD28-

like molecules, taking into account key regions crucial for the functionalities and functional differences between these molecules across various species, including mammals and zebrafish.

Identification of zebrafish Ctla-4: We identified zebrafish Ctla-4 as a homolog of mammalian CTLA-4 based on key conserved structural and functional characteristics. Structurally, the Ctla-4 gene shares similar exon organization compared to mammalian CTLA-4. Ctla-4 is a type I transmembrane protein with typical immunoglobulin superfamily features. Multiple amino acid sequence alignments revealed that Ctla-4 contains a ¹¹³LFPPPY¹¹⁸ motif and a ¹²³GNGT¹²⁶ motif in the ectodomain, and a tyrosine-based ²⁰⁶YVKF²⁰⁹ motif in the distal C-terminal region. These motifs closely resemble MYPPPY, GNGT, and YVKM motifs in mammalian CTLA-4s, which are essential for binding to CD80/CD86 ligands and molecular internalization and signaling inhibition. Despite only 23.7% sequence identity to human CTLA-4, zebrafish Ctla-4 exhibits a similar tertiary structure with a two-layer β -sandwich architecture in its extracellular IgV-like domain. Four cysteine residues responsible for the formation of two pairs of disulfide bonds (Cys²⁰-Cys⁹¹/Cys⁴⁶-Cys⁶⁵ in zebrafish and Cys²¹-Cys⁹²/Cys⁴⁸-Cys⁶⁶ in humans) that connect the two-layer β -sandwich are conserved. Additionally, a separate cysteine residue (Cys¹²⁰ in zebrafish and Cys¹²⁰ in humans) involved in dimerization is also present, and Western blot analysis under reducing and non-reducing conditions confirmed Ctla-4's dimerization. Phylogenetically, Ctla-4 clusters with other known CTLA-4 homologs from different species with high bootstrap probability, while zebrafish Cd28 groups separately with other CD28s. Functionally, Ctla-4 is predominantly expressed on CD4⁺ T and CD8⁺ T cells in zebrafish. It plays a pivotal inhibitory role in T cell activation by competing with CD28 for binding to CD80/86, as validated through a series of both *in vitro* and *in vivo* assays, including microscale thermophoresis assays which demonstrated that Ctla-4 exhibits a significantly higher affinity for Cd80/86 than Cd28 (KD = 0.50 $\pm$ 0.25 μ M vs. KD = 2.64 $\pm$ 0.45 μ M). These findings confirm Ctla-4 as an immune checkpoint molecule, reinforcing its identification within the CTLA-4 family.

Comparison between zebrafish Cd28 and "Cd28l": Zebrafish Cd28 contains an extracellular SYPPPF motif and an intracellular FYIQ motif. The extracellular SYPPPF motif is essential for binding to Cd80/CD86, while the intracellular FYIQ motif likely mediates kinase recruitment and co-stimulatory signaling. In contrast, the "Cd28l" molecule lacks the SYPPPF motif, which is critical for Cd80/CD86 binding, and exhibits strong similarity in its C-terminal 79 amino acids to Ctla-4 rather than Cd28. Consequently, "Cd28l" resembles an atypical Ctla-4-like molecule but fails to exhibit Cd80/CD86 binding activity.

We have incorporated the relevant analysis results into the main text of the revised manuscript and updated Supplementary Figure 1. Additionally, we provide key supplementary analyses here for the reviewer's convenience.

Author response image 1.

Illustrates the alignment of Ctla-4 (XP_005167576.1) and Ctla-4-like (XP_005167567.1, previously referred to as "Cd28l") in zebrafish, generated using ClustalX and Jalview. Conserved and partially conserved amino acid residues are highlighted in color gradients ranging from carnation to red, respectively. The B7-binding motif is encircled with a red square.

Ctla-4/1-209	1	- - M I L L I F I Y P G - - - F A L V Y Q Y R V G K - - - E S I H S F R S K L R P E E V Q V S Y K A L H H E R I C S A V N L S P Y F A D P V R R G N I N	87
Ctla-4-like/1-205	1	M P D I F L A S F S M H N A H C I W D Y E E L N H L N D S S N V T E M I L G N E N - - - - - F S E D K L K M N G K Y C E V D K G E S H - Q N S L H Y E	84
Ctla-4/1-209	88	F G G V D L I A G L R G E D T D V Y R C E E I F P P Y L R A L N G T V V Y I D E I P N C L T A H N A S T Q S Q R G S K S Q T S Y R E T V E N D I O P L L L L Y I L I T S C S	180
Ctla-4-like/1-205	85	D K K F M F K N N P K D I H N S T F S C E S K K K F V E T K A - P E A K L F E I P N C L T A H N A S T Q S Q R G S K S Q T S D R E T V E N D I O P L L L L Y I L I I S C S	176
Ctla-4/1-209	181	F P L Q I A I C K W R T S S S M A P H L S Q K E S Y V K F	209
Ctla-4-like/1-205	177	F P L Q I A I C K W R T S S S M A P H L S Q K E S Y V K F	205

B7-binding motif

(b) Clearer description in the main text of such an analysis to better establish that the mutated gene is a homolog of ctla-4, NOT cd28.

We appreciate the reviewer's advice. Additional confirmation of zebrafish Ctla-4 is detailed in lines 119-126 of the revised manuscript.

(c) Are there mammalian anti-ctla-4 and/or anti-cd28 antibodies that are expected to bind to these zebrafish proteins? If so, looking to see whether staining is lost (or western blotting is lost) in your mutants could be additionally informative. (Our understanding is that your mouse anti-Ctla-4 antibody is raised against recombinant protein generated from this same locus, and so is an elegant demonstration that your mutant eliminates the production of the protein, but unfortunately does not contribute additional information to help establish its homology to mammalian proteins).

This suggestion holds significant value. However, a major challenge in fish immunology research is the limited availability of antibodies suitable for use in fish species; antibodies developed for mammals are generally not applicable. We attempted to use human and mouse anti-CTLA-4 and anti-CD28 antibodies to identify Ctla-4 and Cd28 in zebrafish, but the results were inconclusive, with no expected signals. This outcome likely arises from the low sequence identity between human/mouse CTLA-4 and CD28 and their zebrafish homologs (ranging from 21.3% to 23.7% for CTLA-4 and 21.2% to 24.0% for CD28). Therefore, developing specific antibodies against zebrafish Ctla-4 is essential for advancing this research.

The methods section is generally insufficient and doesn't describe many of the experiments performed in this manuscript. Some examples:

- (a) No description of antibodies used for staining or Western blots (Figure 1C, 1D, 1F).*
- (b) No description of immunofluorescence protocol (Figure 1D, 1F).*
- (c) No description of Western blot protocol (Figure 1C, 2C).*
- (d) No description of electron microscopy approach (Figure 2K).*
- (e) No description of the approach for determining microbial diversity (Entirety of Figure 6).*
- (f) No description of PHA/CFSE/Flow experiments (Figure 7A-E).*
- (g) No description of AlphaFold approach (Figures 7F-G).*
- (h) No description of co-IP approach (Figure 7H).*
- (i) No description of MST assay or experiment (Figure 7I).*
- (j) No description of purification of recombinant proteins, generation of anti-Ctla-4 antibody, or molecular interaction assays (Figures S2 and S6).*

We apologize for this oversight. The methods section was inadvertently incomplete due to an error during the file upload process at submission. This issue has been addressed in the revised manuscript. We appreciate your understanding.

Figure 5 suggests that there are more Th2 cells 1, Th2 cells 2, and NKT cells in ctla-4 mutants through scRNA-seq. However, as the cell numbers for these are low in both genotypes, there is only a single replicate for each genotype scRNA-seq experiment, and dissociation stress can skew cell-type proportions, this finding would be much more

convincing if another method that does not depend on dissociation was used to verify these results. Furthermore, while Th2 cells 2 are almost absent in WT scRNA-seq, KEGG analysis suggests that a major contributor to their clustering may be ribosomal genes (Fig. 5I). Since no batch correction was described in the methods, it would be beneficial to verify the presence of this cluster in ctla-4 mutants and WT animals through other means, such as in situ hybridization or transgenic lines.

We are grateful for the insightful comments provided by the reviewer. Given that research on T cell subpopulations in fish is still in its nascent stages, the availability of specific marker antibodies and relevant transgenic strains remains limited. Our single-cell RNA sequencing (scRNA-seq) analysis revealed that a distinct Th2 subset 2 was predominantly observed in Ctla-4 mutants but was rare in wild-type zebrafish, it suggests that this subset may primarily arise under pathological conditions associated with Ctla-4 mutation. Due to the near absence of Th2 subset 2 in wild-type samples, KEGG enrichment analysis was performed exclusively on this subset from Ctla-4-deficient intestines. The ribosome pathway was significantly enriched, suggesting that these cells may be activated to fulfill their effector functions. However, confirming the presence of Th2 subset 2 using in situ hybridization or transgenic zebrafish lines is currently challenging due to the lack of lineage-specific markers for detailed classification of Th2 cell subsets and the preliminary nature of scRNA-seq predictions.

To address the reviewers' suggestion to confirm compositional changes in Th2 and NKT cells using dissociation-independent methods, we quantified mRNA levels of Th2 (il4, il13, and gata3) and NKT (nkl.2, nkl.4, and prf1.1) cell marker genes via RT-qPCR in intestines from wild-type and mutant zebrafish. As shown in Figure S7B and S7C, these markers were significantly upregulated in Ctla-4-deficient intestines compared to wild-type controls. This indicates an overall increase in Th2 and NKT cell activity in mutant zebrafish, aligning with our scRNA-seq analysis and supports the validity of our initial findings.

Before analyzing the scRNA-seq data, we performed batch correction using the Harmony algorithm via cloud-based Cumulus v1.0 on the aggregated gene-count matrices. This methodological detail has been included in the "Materials and Methods" section of the revised manuscript. Moreover, the RT-qPCR results are presented in Supplementary Figures S7B and S7C.

Quality control (e.g., no. of UMIs, no. of genes, etc.) metrics of the scRNAseq experiments should be presented in the supplementary information for each sample to help support that observed differential expression is not merely an outcome of different sequencing depths of the two samples.

As illustrated in Fig. S5, the quality control data have been supplemented to include the effective cell number of the sample, along with pre- and post-filtering metrics such as nFeature_RNA, nCount_RNA and mitochondrial percentage (percent.mito). Furthermore, scatter plots comparing the basic information of the sample cells before and after filtering are provided.

Some references to prior research lack citations. Examples:

(a)"Given that Ctla-4 is primarily expressed on T cells (Figure 1E-F), and its absence has been shown to result in intestinal immune dysregulation, indicating a crucial role of this molecule as a conserved immune checkpoint in T cell inhibition."

The references were incorporated into line 71 of the revised manuscript.

(b) Line 83: Cite evidence/review for the high degree of conservation in adaptive immunity.

The references were incorporated into line 93 of the revised manuscript.

| (c) Lines 100-102: Cite the evidence that MYPPPY is a CD80/86 binding motif.

The references were incorporated into line 117 of the revised manuscript.

| The text associated with Figure 8 (Lines 280-289) does not clearly state that rescue experiments are being done in mutant zebrafish.

We have provided a clear explanation of the rescue experiments conducted in Ctla-4-deficient zebrafish. This revision has been incorporated into line 319.

| Line 102: Is there evidence from other animals that LFPPPY can function as a binding site for CD80/CD86? Does CD28 also have this same motif?

The extracellular domains of CTLA-4 and CD28, which bind to CD80/CD86, are largely conserved across various species. This conservation is exemplified by a central PPP core motif, although the flanking amino acids exhibit slight variations. In mammals, both CTLA-4 and CD28 feature the conserved MYPPPY motif. By contrast, in teleost fish, such as rainbow trout, CTLA-4 contains an LYPPPY motif, while CD28 has an MYPPPI motif (Ref. 1). Grass carp CTLA-4 displays an LFPPPY motif, whereas its CD28 variant bears an IYPPPF motif. Yeast two-hybrid assays confirm that these motifs facilitate interactions between grass carp CTLA-4 and CD28 with CD80/CD86 (Ref. 2). Similarly, zebrafish Ctla-4 contains the LFPPPY motif observed in grass carp, while Cd28 exhibits a closely related SYPPPF motif.

References:

(1) Bernard, D et al. (2006) Costimulatory Receptors in a Teleost Fish: Typical CD28, Elusive CTLA-4. J Immunol. 176: 4191-4200.

(2) Lu T Z et al. (2022) Molecular and Functional Analyses of the Primordial Costimulatory Molecule CD80/86 and Its Receptors CD28 and CD152 (CTLA-4) in a Teleost Fish. Frontiers in Immunology. 13:885005.

| Line 110-111: Suggest adding citation of these previously published scRNAseq data to the main text in addition to the current description in the Figure legend.

The reference has been added in line 129 in the main text.

| Figure 3B: It would be helpful to label a few of the top differentially expressed genes in Panel B?

The top differentially expressed genes have been labeled in Figure 3B.

| Figure 3G: It's unclear how this analysis was conducted, what this figure is supposed to demonstrate, and in its current form it is illegible.

Figure 3G displays a protein-protein interaction network constructed from differentially expressed genes. The densely connected nodes, representing physical interactions among proteins, provide valuable insights for basic scientific inquiry and biological or biomedical applications. As proteins are crucial to diverse biological functions, their interactions illuminate the molecular and cellular mechanisms that govern both healthy and diseased states in organisms. Consequently, these networks facilitate the understanding of pathogenic and physiological processes involved in disease onset and progression.

To construct this network, we first utilized the STRING database (<https://string-db.org>) to generate an initial network diagram using the differentially expressed genes. This diagram was subsequently imported into Cytoscape (version 3.9.1) for visualization and further analysis. Node size and color intensity reflect the density of interactions, indicating the relative importance of each protein. Figure 3G illustrates that IL1 β was a central cytokine hub in the disease process of intestinal inflammation in Ctla-4-deficient zebrafish.

Expression scale labeling:

(a) Most gene expression scales are not clearly labeled: do they represent mean expression or scaled expression? Has the expression been log-transformed, and if so, which log (natural log? Log10? Log2?). See: Figure 3E, 3I, 4D, 4E, 5B, 5G, 5H, 6I.

The gene expression scales are detailed in the figure legends. Specifically, Figures 3E, 3I, and 6I present heatmaps depicting row-scaled expression levels for the corresponding genes. In contrast, Figures 4D and 4E display heatmaps illustrating the mean expression of these genes. Additionally, the dot plots in Figures 5B, 5G, and 5H visualize the mean expression levels of the respective genes.

(b) For some plots, diverging color schemes (i.e. with white/yellow in the middle) are used for non-diverging scales and would be better represented with a sequential color scale. See: 4D, 4E, and potentially others (not fully clear because of the previous point).

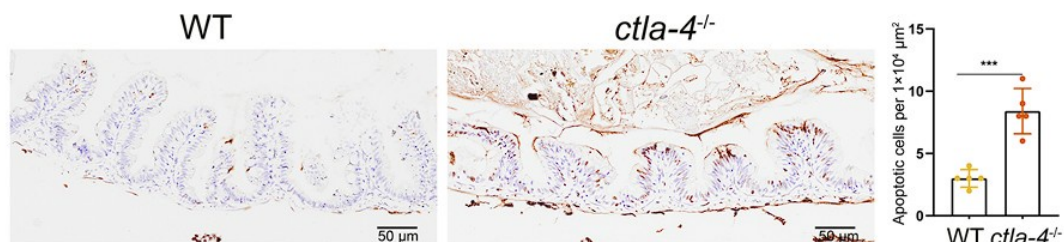
The color schemes in Figures 4D and 4E have been updated to a sequential color scale. The gene expression data depicted in these figures represent mean expression values and have not undergone log transformation. This information has been incorporated into the figure legend for clarity.

Lines 186-187: Though it is merely suggested, apoptotic gene expression can be upregulated as part of the dissociation process for single-cell RNAseq. This would be much stronger if supported by a staining, such as anti-Caspase 3.

Following the reviewer's insightful recommendations, we conducted a TUNEL assay to evaluate apoptosis in the posterior intestinal epithelial cells of both wild-type and Ctla-4-deficient zebrafish. As expected, our results demonstrate a significant increase in epithelial cell apoptosis in Ctla-4-deficient zebrafish compared with wild-type fish. The corresponding data are presented in Figure S6D and have been incorporated into the manuscript. Detailed protocols for the TUNEL assay have also been included in the Materials and Methods section.

Author response image 2.

Illustrates the quantification of TUNEL-positive cells per $1 \times 10^4 \mu\text{m}^2$ in the posterior intestines of both wild-type (WT) and *ctla-4*^{-/-} zebrafish (n = 5). The data demonstrate a comparative analysis of apoptotic cell density between the two genotypes.



Lines 248-251: This manuscript demonstrates gut inflammation and also changes in microbial diversity, but I don't think it demonstrates an association between them, which would require an experiment that for instance rescues one of these changes and shows that it ameliorates the other change, despite still being a ctla-4 mutant.

We appreciate the valuable comments from the reviewer. Recently, the relationship between inflammatory bowel disease (IBD) and gut microbial diversity has garnered considerable attention, with several key findings emerging from human IBD studies. For instance, patients with IBD (including ulcerative colitis and Crohn's disease) exhibit reduced microbial diversity, which is correlated with disease severity. This decrease in microbial richness is thought to stem from the loss of normal anaerobic bacteria, such as *Bacteroides*, *Eubacterium*, and *Lactobacillus* (Refs. 1-6). Research using mouse models has shown that inflammation increases oxygen and nitrate levels within the intestinal lumen, along with elevated host-derived electron acceptors, thereby promoting anaerobic respiration and overgrowth of *Enterobacteriaceae* (Ref 7). Consistent with these findings, our study observed a significant enrichment of *Enterobacteriaceae* in the inflamed intestines of Ctla-4-deficient zebrafish, which supporting the observations in mice. Despite this progress, the zebrafish model for intestinal inflammation remains under development, with limitations in available techniques for manipulating intestinal inflammation and reconstructing gut microbiota. These challenges hinder investigations into the association between intestinal inflammation and changes in microbial diversity. We plan to address these issues through ongoing technological advancements and further research. We thank the reviewer for their understanding.

References:

- (1) Ott S J, Musfeldt M, Wenderoth D F, Hampe J, Brant O, Fölsch U R *et al.* (2004) Reduction in diversity of the colonic mucosa associated bacterial microflora in patients with active inflammatory bowel disease. *Gut* 53:685-693.
- (2) Manichanh C, Rigottier-Gois L, Bonnaud E, Gloux K, Pelletier E, Frangeul L *et al.* (2006) Reduced diversity of faecal microbiota in Crohn's disease revealed by a metagenomic approach. *Gut* 55:205-211.
- (3) Qin J J, Li R Q, Raes J, Arumugam M, Burgdorf K S, Manichanh C *et al.* (2010) A human gut microbial gene catalogue established by metagenomic sequencing. *Nature* 464:59-U70.
- (4) Sha S M, Xu B, Wang X, Zhang Y G, Wang H H, Kong X Y *et al.* (2013) The biodiversity and composition of the dominant fecal microbiota in patients with inflammatory bowel disease. *Diagn Micr Infec Dis* 75:245-251.
- (5) Ray K. (2015) IBD. Gut microbiota in IBD goes viral. *Nat Rev Gastroenterol Hepatol* 12:122.
- (6) Papa E, Docktor M, Smillie C, Weber S, Preheim S P, Gevers D *et al.* (2012) Non-Invasive Mapping of the Gastrointestinal Microbiota Identifies Children with Inflammatory Bowel Disease. *Plos One* 7: e39242-39254.
- (7) Hughes E R, Winter M G, Duerkop B A, Spiga L, de Carvalho T F, Zhu W H *et al.* (2017) Microbial Respiration and Formate Oxidation as Metabolic Signatures of Inflammation-Associated Dysbiosis. *Cell Host Microbe* 21:208-219.

Lines 270-272 say that interaction between Cd28/ctla-4 and Cd80/86 was demonstrated through bioinformatics, flow-cytometry, and Co-IP. Does this need to reference Fig S6D for the flow data? Figures 7F-G are very hard to read or comprehend as they are very small. Figure 7H is the most compelling evidence of this interaction and might stand out better if emphasized with a sentence referencing it on its own in the manuscript.

In this study, we utilized an integrated approach combining bioinformatics prediction, flow cytometry, and co-immunoprecipitation (Co-IP) to comprehensively investigate and validate the interactions between Cd28/Ctla-4 and Cd80/86. Flow cytometry analysis, as depicted in Supplementary Figure 6D (revised as Supplementary Figure 8F), demonstrated the surface expression of Cd80/86 on HEK293T cells and quantified their interactions with Cd28 and Ctla-4. These experiments not only validated the interactions between Cd80/86 and Cd28/Ctla-4 but also revealed a dose-dependent relationship, providing robust supplementary evidence for the molecular interactions under investigation. Furthermore, in Figure 7F-G, the axis font sizes were enlarged to improve readability. Additionally, in response to reviewers' feedback, we have emphasized Figure 7H, which presents the most compelling evidence for molecular interactions, by including a standalone sentence in the text to enhance its prominence.

For Figure 7A-E, for non-immunologists, it is unclear what experiment was performed here - it would be helpful to add a 1-sentence summary of the assay to the main text or figure legend.

We apologize for this oversight. Figures 7A–E illustrate the functional assessment of the inhibitory role of Ctla-4 in Cd80/86 and Cd28-mediated T cell activation. A detailed description of the methodologies associated with Figures 7A–E is provided in the ‘Materials and Methods’ section of the revised manuscript.

For Figure 7F-G, it is extremely hard to read the heat map legends and the X and Y-axis. Also, what the heatmaps show and how that fits the overall narrative can be elaborated significantly.

We regret this oversight. To enhance clarity, we have increased the font size of the heatmap legends and the X and Y-axes, as shown in the following figure. Additionally, a detailed analysis of these figures is provided in lines 299–306 of the main text.

In general, the main text that accompanies Figure 7 should be expanded to more clearly describe these experiments/analyses and their results.

We have conducted a detailed analysis of the experiments and results presented in Figure 7. This analysis is described in lines 278-314.

Reviewer #2:

The scRNASeq assay is missing some basic characterization: how many WT and mutant fish were assayed in the experiment? how many WT and mutant cells were subject to sequencing? Before going to the immune cell types, are intestinal cell types comparable between the two conditions? Are there specific regions in the tSNE plot in Figure 4A abundant of WT or ctla-4 mutant cells?

In the experiment, we analyzed 30 wild-type and 30 mutant zebrafish for scRNA-seq, with an initial dataset comprising 8,047 cells in the wild-type group and 8,321 cells in the mutant group. Sample preparation details are provided on lines 620-652. Due to the relatively high expression of mitochondrial genes in intestinal tissue, quality control filtering yielded 3,263 cells in the wild-type group and 4,276 cells in the mutant group. Given that the intestinal tissues were dissociated using identical protocols, the resulting cell types are comparable between the two conditions. Both the wild-type and Ctla-4-deficient groups contained enterocytes, enteroendocrine cells, smooth muscle cells, neutrophils, macrophages, B cells, and a cluster of T/NK/ILC-like cells. Notably, no distinct regions were enriched for either condition in the tSNE plot (Figure 4A).

The cell proliferation experiment using PHA stimulation assay demonstrated the role of Ctla-4 in cell proliferation, while the transcriptomic evidence points towards activation rather than an overall expansion of T-cell numbers. This should be discussed towards a more comprehensive model of how subtypes of cells can be differentially proliferating in the disease model.

In the PHA-stimulated T cell proliferation assay, we aimed to investigate the regulatory roles of Ctla-4, Cd28, and Cd80/86 in T cell activation, focusing on validating Ctla-4's inhibitory function as an immune checkpoint. While our study examined general regulatory mechanisms, it did not specifically address the distinct roles of Ctla-4 in different T cell subsets. We appreciate the reviewer's suggestion to develop a more comprehensive model that elucidates differential T cell activation across various subsets in disease models. However, due to the nascent stage of research on fish T cell subsets and limitations in lineage-specific antibodies and transgenic strains, such investigations are currently challenging. We plan to pursue these studies in the future. Despite these constraints, our single-cell RNA sequencing data revealed an increased proportion of Th2 subset cells in Ctla-4-deficient zebrafish, as evidenced by elevated expression levels of Th2 markers (Il4, Il13, and Gata3) via RT-qPCR (see Figures S7B). Notably, recent studies in mouse models have shown that naïve T cells from CTLA-4-deficient mice tend to differentiate into Th2 cells post-proliferation, with activated Th2 cells secreting higher levels of cytokines like IL-4, IL-5, and IL-13, thereby exerting their effector functions (Refs. 1-2). Consequently, our findings align with observations in mice, suggesting conserved CTLA-4 functions across species. We have expanded the "Discussion" section to clarify these points.

References:

- (1) Bour-Jordan H, Grogan J L, Tang Q Z, Auger J A, Locksley R M, Bluestone J A et al. (2003) CTLA-4 regulates the requirement for cytokine-induced signals in T_H2 lineage commitment. *Nature Immunology* 4: 182-188.
- (2) Khattri Roli, Auger, Julie A, Griffin Matthew D, Sharpe Arlene H, Bluestone Jeffrey A et al. (1999) Lymphoproliferative Disorder in CTLA-4 Knockout Mice Is Characterized by CD28-Regulated Activation of Th2 Responses. *The Journal of Immunology* 162:5784-5791.

It would be nice if the authors could also demonstrate whether other tissues in the zebrafish have an inflammation response, to show whether the model is specific to IBD.

In addition to intestinal tissues, we also performed histological analysis on the liver of Ctla-4-deficient zebrafish. The results showed that Ctla-4 deficiency led to mild edema in a few hepatocytes, and lymphocyte infiltration was not significant. Compared to the liver, we consider intestinal inflammation to be more pronounced.

Some minor comments on terminology

(a) "multiomics" usually refers to omics experiments with different modalities (e.g. transcriptomics, proteomics, metabolomics etc), while the current paper only has transcriptomics assays. I wouldn't call it "multiomics" analysis.

We appreciate the reviewer's attention to this issue. The "multi-omics" has been revised to "transcriptomics".

(b) In several parts of the figure legend the author mentioned "tSNE nonlinear clustering" (Figures 4A and 5A). tSNE is an embedding method rather than a clustering method.

The "tSNE nonlinear clustering" has been revised to "tSNE embedding".

(c) Figure 1E is a UMAP rather than tSNE.

The "tSNE" has been revised to "UMAP" in the figure legend in line 1043.

Reviewer #3:

Line 28: The link is not directly reflected in this sentence describing CTLA-4 knockout mice.

We appreciate the reviewer for bringing this issue to our attention. We have expanded our description of CTLA-4 knockout mice on lines 77-84.

Line 80-83: There is a lack of details about the CTLA-4-deficient mice. The factor that Th2 response could be induced has been revealed in mouse model. See the reference entitled "CTLA-4 regulates the requirement for cytokine-induced signals in TH2 lineage commitment" published in Nature Immunology.

We thank the reviewer for providing valuable references. We have added descriptions detailing the differentiation of T cells into Th2 cells in CTLA-4-deficient mice on lines 78–81, and the relevant references have been cited in the revised manuscript.

To better introduce the CTLA-4 immunobiology, the paper entitled "Current Understanding of Cytotoxic T Lymphocyte Antigen-4 (CTLA-4) Signaling in T-Cell Biology and Disease Therapy" published in Molecules and Cells should be referred.

We have provided additional details on CTLA-4 immunology (lines 75-84) and have included the relevant reference in the revised manuscript.

In current results, there are many sentences that should be moved to the discussion, such as lines 123-124, lines 152-153, lines 199-200, and lines 206-207. So, the result sections just describe the results, and the discussions should be put together in the discussion.

We have relocated these sentences to the 'Discussion' section and refined the writing.

In the discussion, the zebrafish enteritis model, such as DSS/TNBS and SBMIE models, should also be compared with the current CTLA-4 knockout model. Also, the comparison between the current fish IBD model and the previous mouse model should also be included, to enlighten the usage of CTLA-4 knockout zebrafish IBD model.

We compared the phenotypes of our current CtlA-4-knockout zebrafish IBD model with other models, including DSS-induced IBD models in zebrafish and mice, as well as TNBS- and SBM-induced IBD models in zebrafish. The details are included in the "Discussion" section (lines 353-365).

As to the writing, the structure of the discussion is poor. The paragraphs are very long and hard to follow. Many findings from current results were not yet discussed. I just can't find any discussion about the alteration of intestinal microbiota.

In response to the reviewers' constructive feedback, we have revised and enhanced the discussion section. Furthermore, we have integrated the most recent research findings relevant to this study into the discussion to improve its relevance and comprehensiveness.

In the discussion, the aerobic-related bacteria in 16s rRNA sequencing results should be focused on echoing the histopathological findings, such as the emptier gut of CTLA-4 knockout zebrafish.

As mentioned above, the discussion section has been revised and expanded to provide a better understanding of the potential interplay among intestinal inflammatory pathology, gut microbiota alterations, and immune cell dysregulation in Ctla-4-deficient zebrafish. Furthermore, promising avenues for future research that warrant further investigation were also discussed.

In the current method, there are no descriptions for many used methods, which already generated results, such as WB, MLR, MST, Co-IP, AlphaFold2 prediction, and how to make currently used anti-zfCTLA4 antibody. Also, there is a lack of description of the method of the husbandry of knockout zebrafish line.

We regret these flaws. The methods section was inadvertently incomplete due to an error during the file upload process at submission. This issue has been rectified in the revised manuscript. Additionally, Ctla-4-deficient zebrafish were reared under the same conditions as wild-type zebrafish, and the rearing methods are now described in the "Generation of Ctla-4-deficient zebrafish" section of the Materials and Methods.

Line 360: the experimental zebrafish with different ages could be a risk for unstable intestinal health. See the reference entitled "The immunoregulatory role of fish-specific type II SOCS via inhibiting metaflammation in the gut-liver axis" published in Water Biology and Security. The age-related differences in zebrafish could be observed in the gut.

We appreciate the reviewers' reminders. The Ctla-4 mutant zebrafish used in our experiments were 4 months old, while the wild-type zebrafish ranged from 4 to 6 months old. These experimental fish were relatively young and uniformly distributed in age. During our study, we examined the morphological structures of the intestines in zebrafish aged 4 to 6 months and observed no significant abnormalities. These findings align with previous research indicating no significant difference in intestinal health between 3-month-old and 6-month-old wild-type zebrafish (Ref. 1). Consequently, we conclude that there is no notable aging-related change in the intestines of zebrafish aged 4 to 6 months. This reduces the risk associated with age-related variables in our study. We have added an explanation stating that the Ctla-4 mutant zebrafish used in the experiments were 4 months old (Line 449) in the revised manuscript.

Reference

(1) Shan Junwei, Wang Guangxin, Li Heng, Zhao Xuyang et al. (2023) The immunoregulatory role of fish-specific type II SOCS via inhibiting metaflammation in the gut-liver axis. *Water Biology and Security* 2: 100131-100144.

Section "Generation of Ctla-4-deficient zebrafish": There is a lack of description of PCR condition for the genotyping.

The target DNA sequence was amplified at 94 °C for 4 min, followed by 35 cycles at 94°C for 30 s, 58°C for 30 s and 72°C for 30 s, culminating in a final extension at 72 °C for 10 min. The polymerase chain reaction (PCR) conditions are described in lines 458-460.

How old of the used mutant fish? There should be a section "sampling" to provide the sampling details.

The "Sampling" information has been incorporated into the "Materials and Methods" section of the revised manuscript. Wild-type and Ctla-4-deficient zebrafish of varying months were housed in separate tanks, each labeled with its corresponding birth date. Experiments utilized Ctla-4-deficient zebrafish aged 4 months and wild-type zebrafish aged between 4 to 6 months.

Line 378-380: The index for the histopathological analysis should be detailed, rather than just provide a reference. I don't think these indexes are good enough to specifically describe the pathological changes of intestinal villi and mucosa. It is suggested to improve with detailed parameters. As described in the paper entitled "Pathology of Gastric Intestinal Metaplasia: Clinical Implications" published in Am J Gastroenterol., histochemical, normal gastric mucins are pH neutral, and they stain magenta with periodic acid-Schiff (PAS). In an inflamed gut, acid mucins replace the original gastric mucins and are stained blue with Alcian blue (AB). So, to reveal the pathological changes of goblet cells and involved mucin components, AB staining should be added. Also, for the number of goblet cells in the inflammatory intestine, combining PAS and AB staining is the best way to reveal all the goblet cells. In Figure 2, there were very few goblet cells. The infiltration of lymphocytes and the empty intestinal lumen could be observed. Thus, the ratio between the length of intestinal villi and the intestinal ring radius should be calculated.

In response to the reviewers' valuable suggestions, we have augmented the manuscript by providing additional parameters related to the pathological changes observed in the Ctla-4-deficient zebrafish intestines, including the mucin component changes identified through PAS and AB-PAS staining, the variations in the number of goblet cells evaluated by AB-PAS staining, and the ratio of intestinal villi length to the intestinal ring radius, as illustrated in the following figures. These new findings are detailed in the "Materials and Methods" (lines 563-566) and "Results" (lines 143-146) sections, along with Supplementary Figure S3 of the revised manuscript.

Section "Quantitative real-time PCR": What's the machine used for qPCR? How about the qPCR validation of RNA seq data? I did not see any related description of data and methods for qPCR validation. In addition, beta-actin is not a stable internal reference gene, to analyze inflammation and immune-related gene expression. See the reference entitled "Actin, a reliable marker of internal control?" published in Clin Chim Acta. Other stable housekeeping genes, such as EF1alpha and 18s, could be better internal references.

RT-qPCR experiments were conducted using a PCR thermocycler device (CFX Connect Real-Time PCR Detection System with Precision Melt Analysis™ Software, Bio-Rad, Cat. No. 1855200EM1). This information has been incorporated into lines 608-610 of the "Materials and Methods" section. In these experiments, key gene sequences of interest, including il13, mpx, and il1β, were extracted from RNA-seq data for RT-qPCR validation. To ensure accurate normalization, potential internal controls were evaluated, and β-actin was identified as a suitable candidate due to its consistent expression levels in the intestines of both wild-type and Ctla-4-deficient zebrafish. The use of β-actin as an internal control is further supported by its application in recent studies on intestinal inflammation (Refs 1–2).

References:

(1) Tang Duo Zhuang, Zeng Ting, Wang Yiting, Cui Hui et al. (2020) Dietary restriction increases protective gut bacteria to rescue lethal methotrexate-induced intestinal toxicity. Gut Microbes 12: 1714401-1714422.

(2) Malik Ankit, Sharma Deepika et al. (2023) Epithelial IFN γ signaling and compartmentalized antigen presentation orchestrate gut immunity. *Nature* 623: 1044-1052.

| *How to generate sCtla-4-Ig, Cd28-Ig and Cd80/86? No method could be found.*

We apologize for the omission of these methods. The detailed protocols have now been added to the "Materials and Methods" section of the revised manuscript (lines 464-481).

| *Figure 5: As reviewed in the paper entitled "Teleost T and NK cell immunity" published in Fish and Shellfish Immunology, two types of NK cell homologues have been described in fish: non-specific cytotoxic cells and NK-like cells. There is no NKT cell identified in the teleost yet. Therefore, "NKT-like" could be better to describe this cell type.*

We refer to "NKT" cells as "NKT-like" cells, as suggested.

| *For the supplementary data of scRNA-seq, there lacks the details of expression level.*

The expression levels of the corresponding genes are provided in Supplemental Table 4.

| *Supplemental Table 1: There are no accession numbers of amplified genes.*

The accession numbers of the amplified genes are included in Supplemental Table 1.

| *The English needs further editing.*

We have made efforts to enhance the English to meet the reviewers' expectations.

| *Line 32: The tense should be the past.*

This tense error has been corrected.

| *Line 363-365: The letter of this approval should be provided as an attachment.*

The approval document is provided as an attachment.

| *Line 376: How to distinguish the different intestinal parts? Were they judged as the first third, second third, and last third parts of the whole intestine?*

The differences among the three segments of zebrafish intestine are apparent. The intestinal tube narrows progressively from the anterior to the mid-intestine and then to the posterior intestine. Moreover, the boundaries between the intestinal segments are well-defined, facilitating the isolation of each segment.

| *Line 404: Which version of Cytoscape was used?*

The version of Cytoscape used in this study is 3.9.1. Information about the Cytoscape version is provided on line 603.

| *The product information of both percoll and cell strainer should be provided.*

The information regarding Percoll and cell strainers has been added on lines 626 and 628, respectively.

| *Line 814: Here should be a full name to tell what is MST.*

The acronym MST stands for "Microscale Thermophoresis", a technique that has been referenced on lines 1157-1158.

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